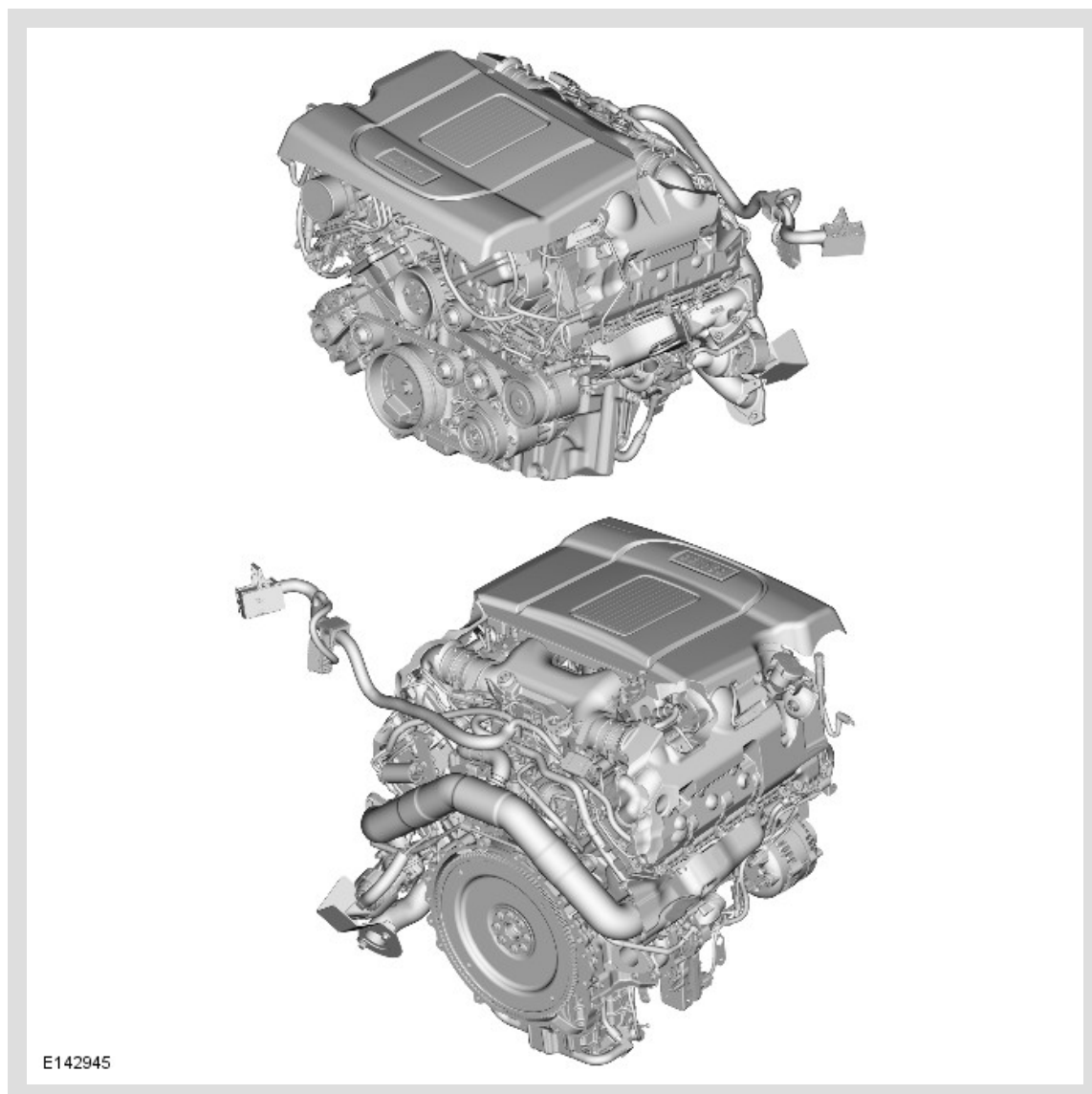


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2015.0 RANGE ROVER (LG), 303-01

# **ENGINE - TDV8 4.4L DIESEL**

## EXTERNAL VIEWS

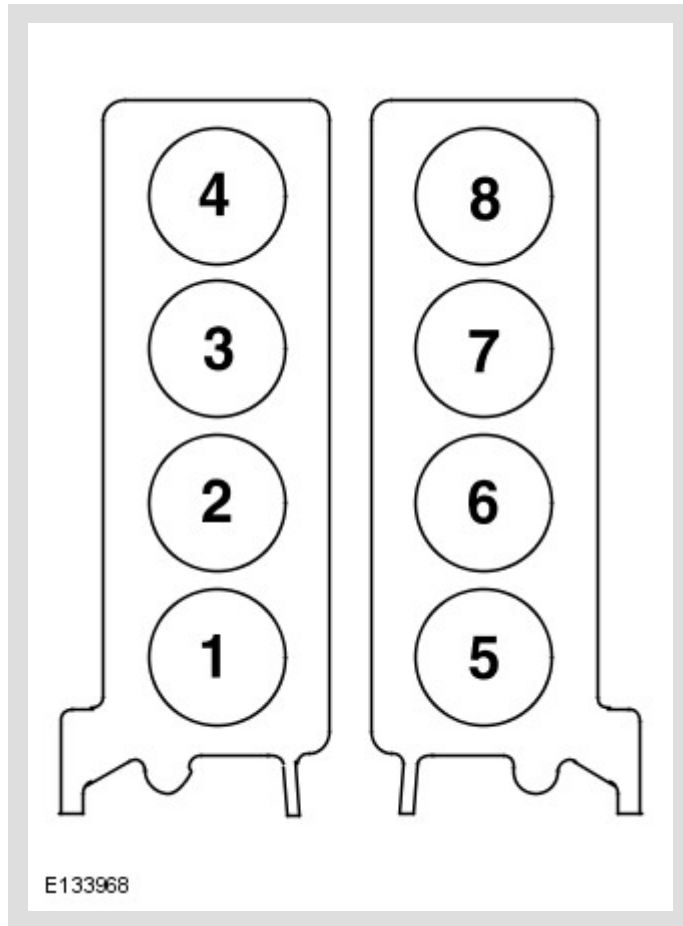


## OVERVIEW

The TDV8 4.4L diesel is a direct injection, 8-cylinder engine with 2 banks of 4 cylinders, arranged at 90 degrees to each other.

### Cylinder Numbering

DIN standard cylinder numbering



**DIN cylinder firing order** 1,5,4,2,6,3,7,8

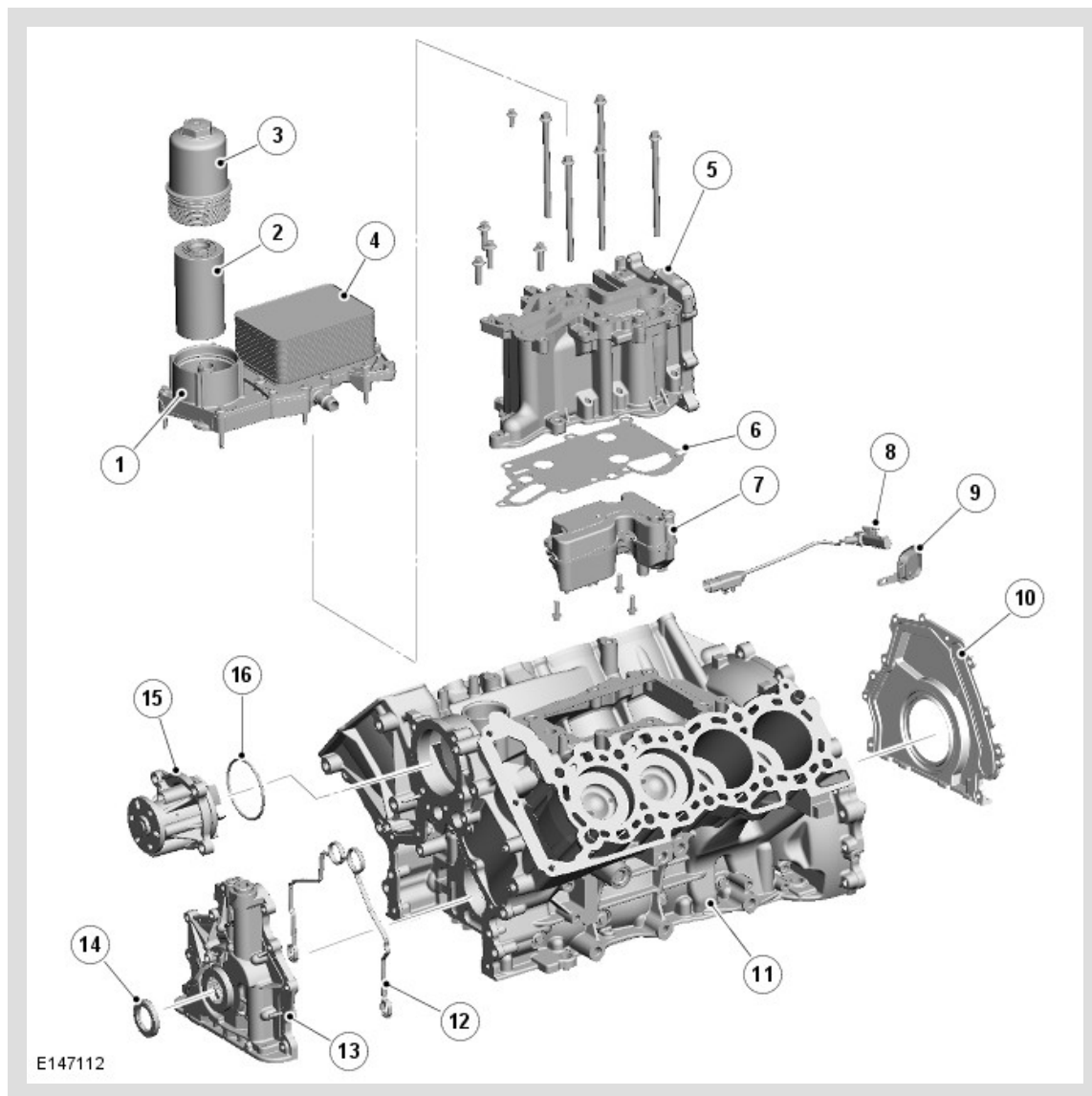
There are 4 valves per cylinder, which are operated by 2 overhead camshafts per cylinder.

The engine is liquid cooled and turbo-charged. The fuel injection system features common rail technology.

The cylinder block is coupled with a separate oil pan to provide a lightweight, compact and very stiff bottom end. The cylinder heads are fitted with integrated intake manifold and camshaft covers. The exhaust manifolds are unique for each cylinder head. Acoustic injector covers and an acoustic top cover are fitted over the upper engine to absorb engine-generated noise.

## DESCRIPTION

### CYLINDER BLOCK COMPONENTS



| ITEM | DESCRIPTION                                  |
|------|----------------------------------------------|
| 1    | Oil filter housing                           |
| 2    | Oil filter element                           |
| 3    | Oil filter cap                               |
| 4    | Oil cooler assembly                          |
| 5    | Exhaust gas recirculation cooler             |
| 6    | EGR cooler gasket                            |
| 7    | Crankcase ventilation oil separator assembly |
|      |                                              |

|    |                               |
|----|-------------------------------|
| 8  | Crankshaft position sensor    |
| 9  | Cylinder block aperture cover |
| 10 | Crankshaft rear oil seal      |
| 11 | Cylinder block                |
| 12 | Oil pump gasket               |
| 13 | Oil pump                      |
| 14 | Crankshaft front oil seal     |
| 15 | Coolant pump                  |
| 16 | Coolant pump seal             |

## CYLINDER BLOCK

The cylinders and crankcase are contained in the closed-deck cylinder block, which has a hollow beam structure and features fully cross-bolted main bearing journals. With this type of construction less material is required than for a conventional cylinder block, therefore reducing engine weight and length.

Lubrication oil is distributed through the cylinder block, via the main oil gallery and channels bored in the block, to all critical moving parts. These channels divert oil to the main and big-end bearings via holes machined into the crankshaft.

A single connection at the right rear of the oil cooler and oil filter assembly connects a pipe to each of the turbochargers via a T-piece connector. Oil is supplied under pressure via this connection, from the oil pump, to provide lubrication for the bearings of the turbochargers.

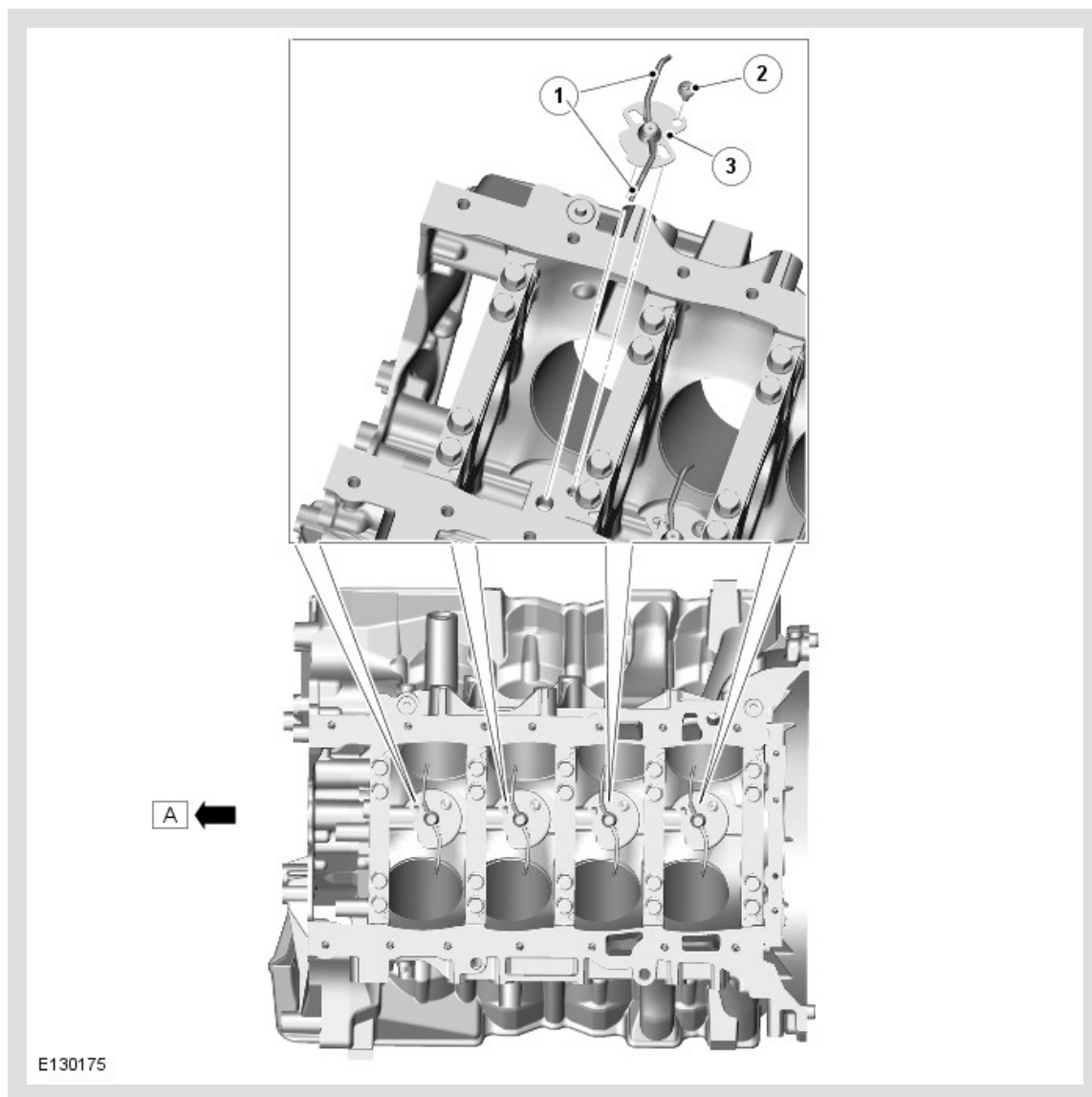
Cylinder cooling is achieved by coolant circulating through chambers in the cylinder block casting.

Two hollow dowels are used to locate the cylinder heads to the cylinder block, one on each side at the rear of the unit.

A port is included at the rear right and left sides of the oil pan, below the turbochargers, to connect the turbocharger oil return pipes to a dedicated reservoir (see TURBOCHARGER OIL RETURN section for more information).

Threaded plugs sealing the lubrication cross-drilling galleries are located at the front left and right sides of the cylinder block. Plugs for the main lubrication gallery are included at the front and rear of the cylinder block.

## PISTON COOLING JETS

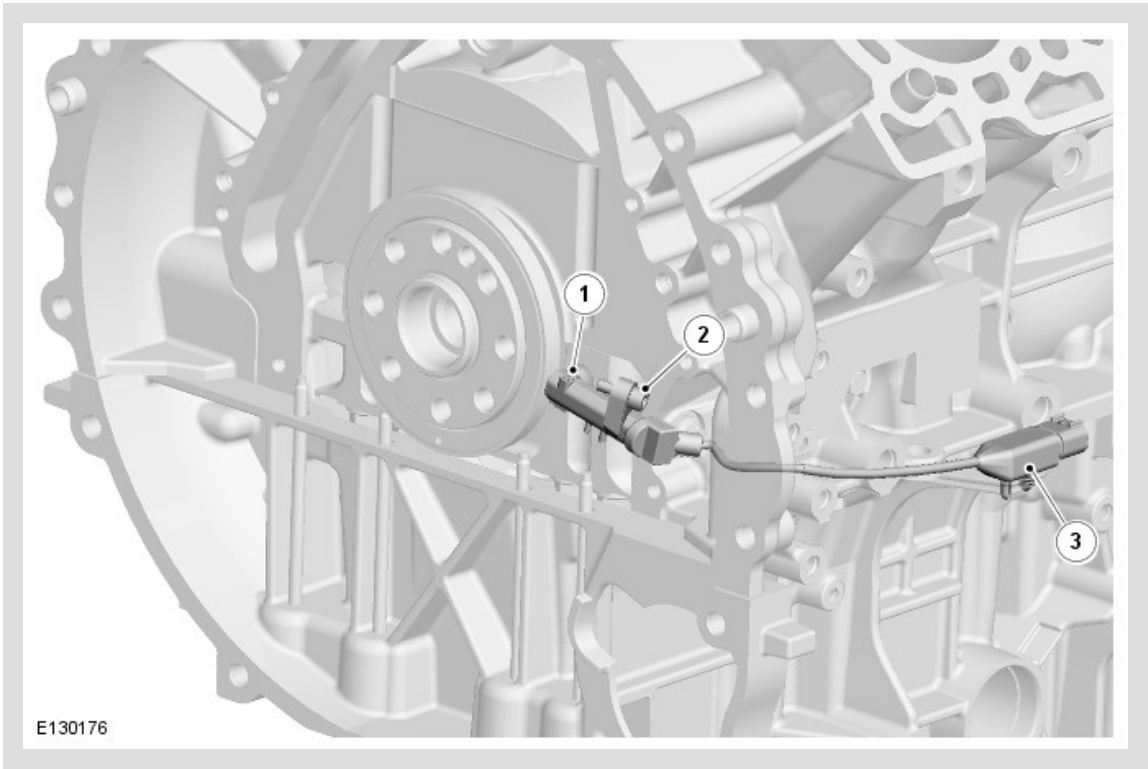


| ITEM | DESCRIPTION         |
|------|---------------------|
| A    | Front of engine     |
| 1    | Piston cooling jets |
| 2    | Screw               |
| 3    | Plate               |

Jets located in the cylinder block provide piston and gudgeon pin lubrication and cooling.

These jets spray oil onto the inside of the piston. The oil then flows through 2 internal wave shaped channels to help cool each piston crown.

CRANKSHAFT POSITION SENSOR



| ITEM | DESCRIPTION                |
|------|----------------------------|
| 1    | Crankshaft position sensor |
| 2    | Bolt                       |
| 3    | Electrical connector       |

The CKP (crankshaft position) sensor is located at the rear of the crankshaft, behind the drive plate in the right side of the rear oil seal retainer. The sensor bolts into the rear oil seal retainer.

The CKP sensor provides an input of engine crankshaft speed and position. The sensor works on the Hall effect principle and scans a reluctor ring on the crankshaft. An air gap of 0.4 to 1.5 mm, between the reluctor ring and the sensor, is achieved by the positional mounting of the sensor. The air gap is not adjustable.

For additional information, refer to: [Electronic Engine Controls](#) (303-14E Electronic Engine Controls - TDV8 4.4L Diesel, Description and Operation).

STARTER MOTOR

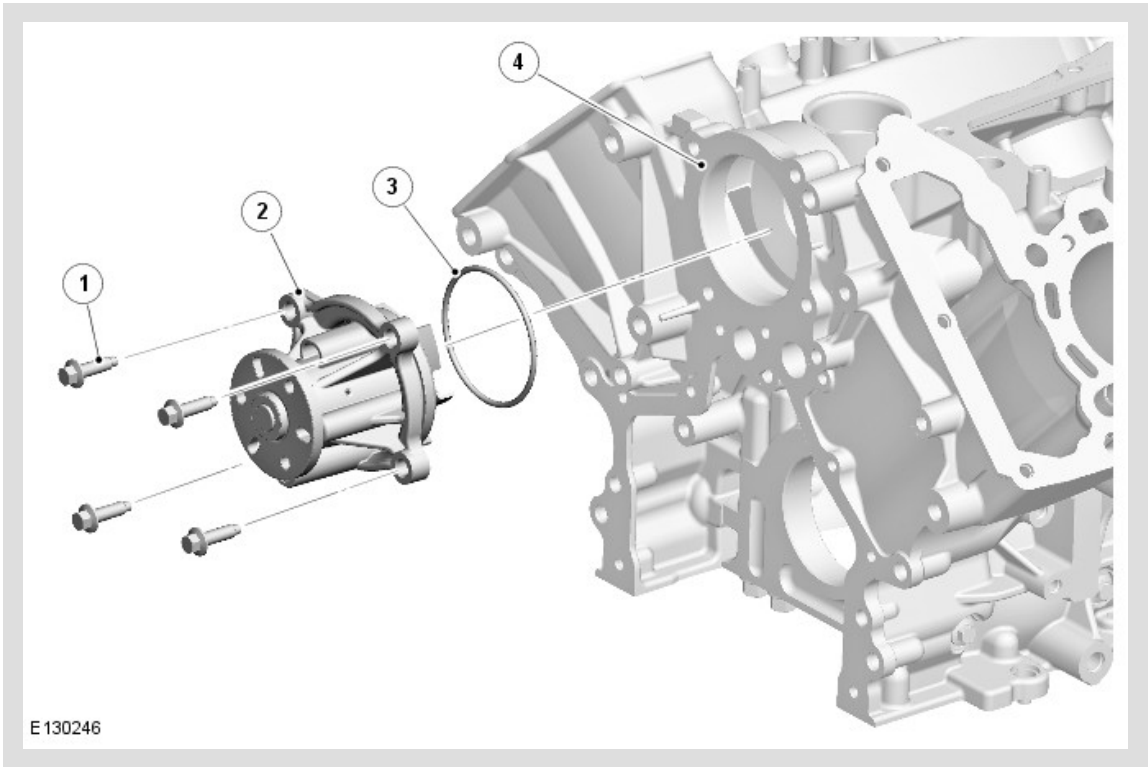
The engine starter motor is installed at the rear right side of the oil pan, at the cylinder block to oil pan split line.

For additional information, refer to: [Starting System](#) (303-06D Starting System - TDV8 4.4L Diesel, Description and Operation).

COOLANT DRAIN PLUG

A coolant drain plug is installed in the cylinder block on the left side below cylinder number 5.

COOLANT PUMP



| ITEM | DESCRIPTION    |
|------|----------------|
| 1    | Bolt (4 off)   |
| 2    | Coolant pump   |
| 3    | O-ring seal    |
| 4    | Cylinder block |

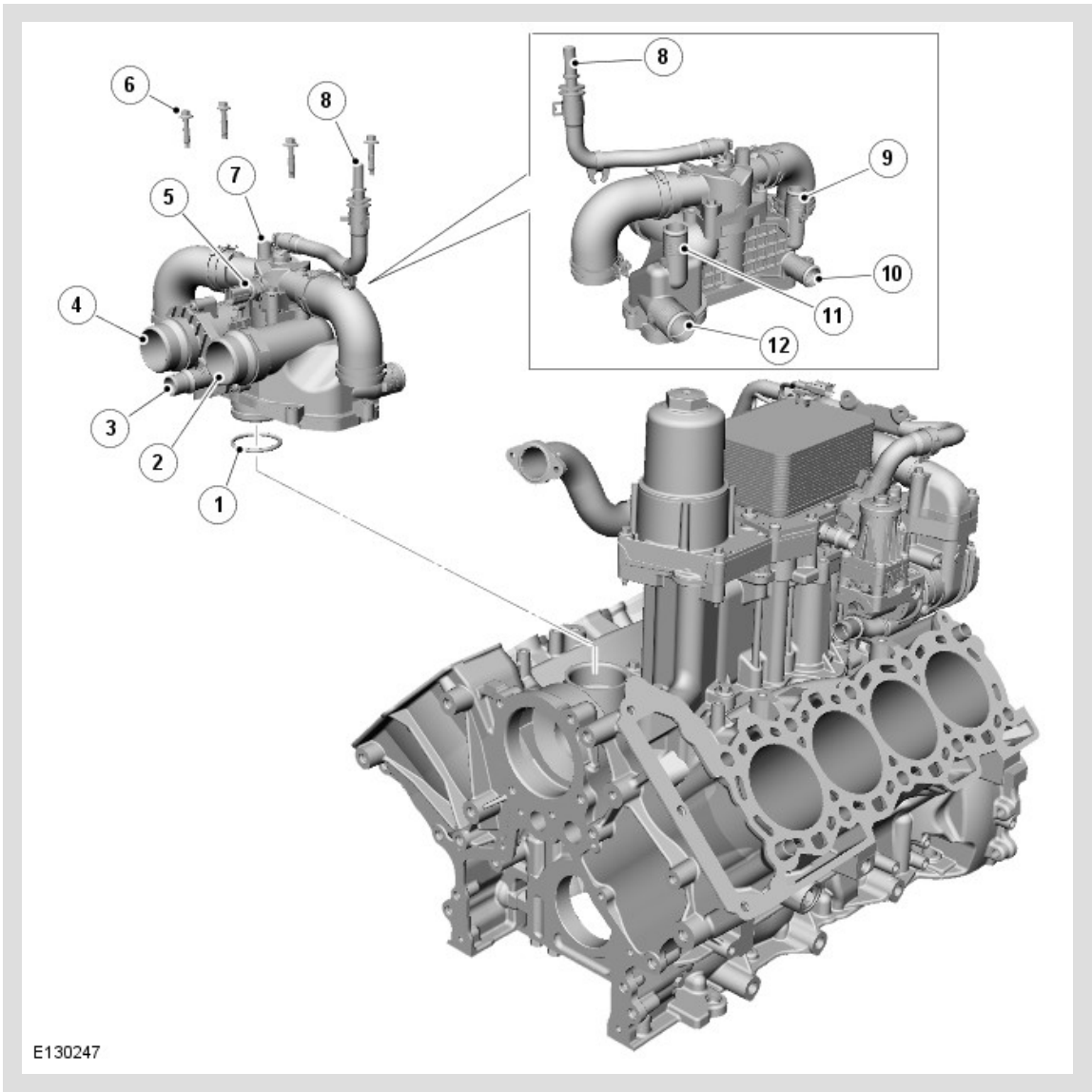
The coolant pump is installed on the center of the cylinder block front face and is secured and sealed using 4 bolts and an O-ring. The accessory drive belt drives the coolant pump.

OIL FILTER AND COOLANT ASSEMBLY

The oil filter and oil cooler assembly is located on top of the EGR (exhaust gas recirculation) cooler, which is located in the center of the 'vee' of the cylinder block.

The oil filter is a replaceable cartridge installed on an adapter.

THERMOSTAT ASSEMBLY



| ITEM | DESCRIPTION                             |
|------|-----------------------------------------|
| 1    | Seal                                    |
| 2    | Connection from radiator upper hose     |
| 3    | Connection to expansion tank            |
| 4    | Thermostat and housing assembly         |
| 5    | Engine Coolant Temperature (ECT) sensor |

|    |                                      |
|----|--------------------------------------|
| 6  | Bolt (4 off)                         |
| 7  | Housing                              |
| 8  | Coolant bleed to radiator upper hose |
| 9  | Coolant supply to heater core        |
| 10 | EGR cooler inlet                     |
| 11 | Coolant return from heater core      |
| 12 | EGR cooler outlet                    |

The thermostat assembly connects the cooling channels of the cylinder block to the cooling channels of the left and right cylinder heads and provides the coolant outlet for the EGR and coolant return.

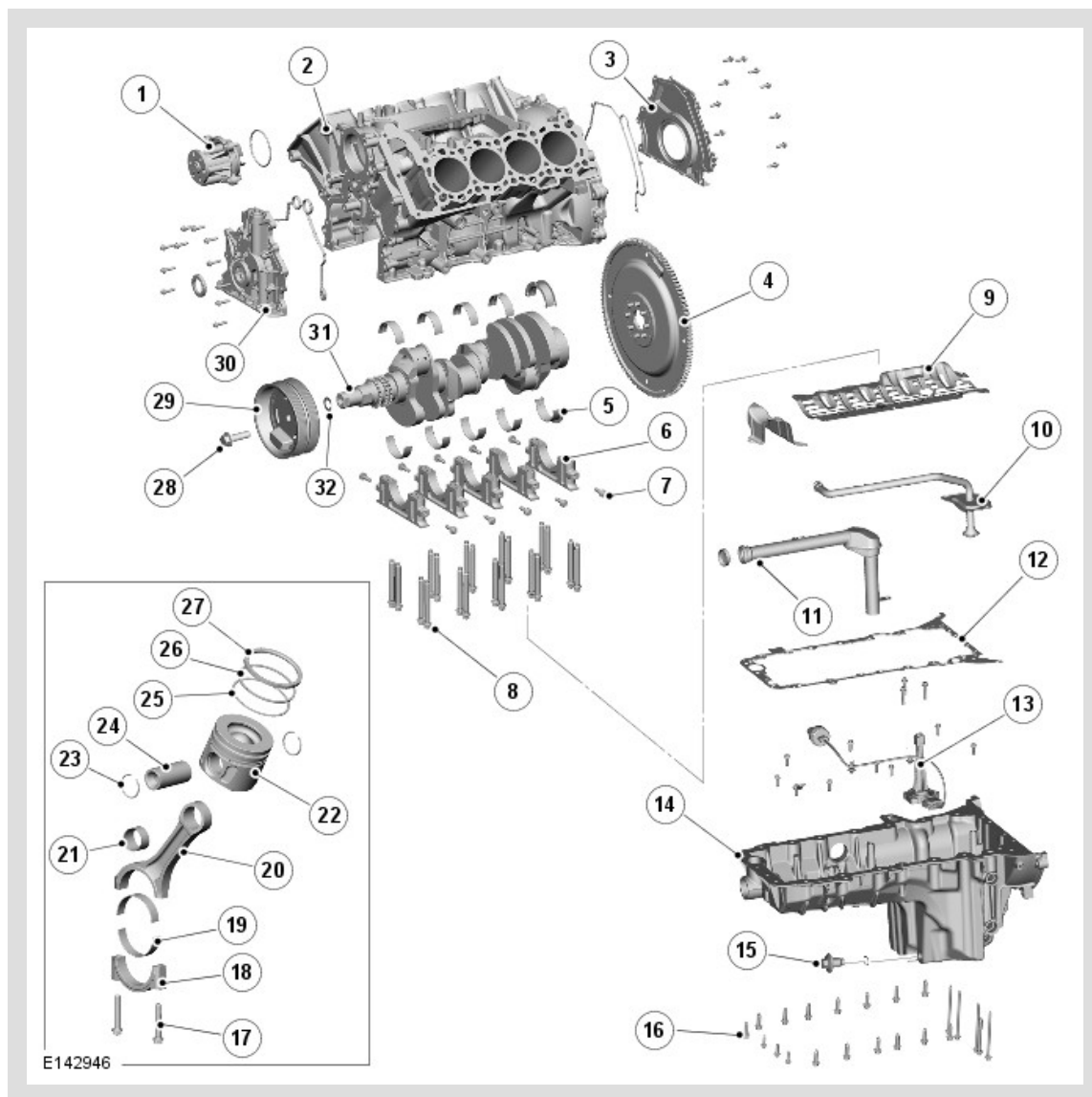
For additional information, refer to: [Engine Cooling](#) (303-03F Engine Cooling - TDV8 4.4 L Diesel, Description and Operation).

On the front of the thermostat assembly are three hose connections for the radiator upper hose, expansion tank and radiator lower hose. A connection from the top of the thermostat assembly connections to the radiator upper hose and three connections at the rear of the assembly provide for connections to the EGR cooler inlet, EGR cooler outlet, heater core supply and heater core return.

An ECT (engine coolant temperature) sensor is located in a hole in the front of the thermostat assembly. The sensor is sealed with an O-ring seal and secured with a spring clip. The ECT sensor is connected to the ECM (engine control module) which uses the coolant temperature information for a number of functions.

For additional information, refer to: [Electronic Engine Controls](#) (303-14E Electronic Engine Controls - TDV8 4.4L Diesel, Description and Operation).

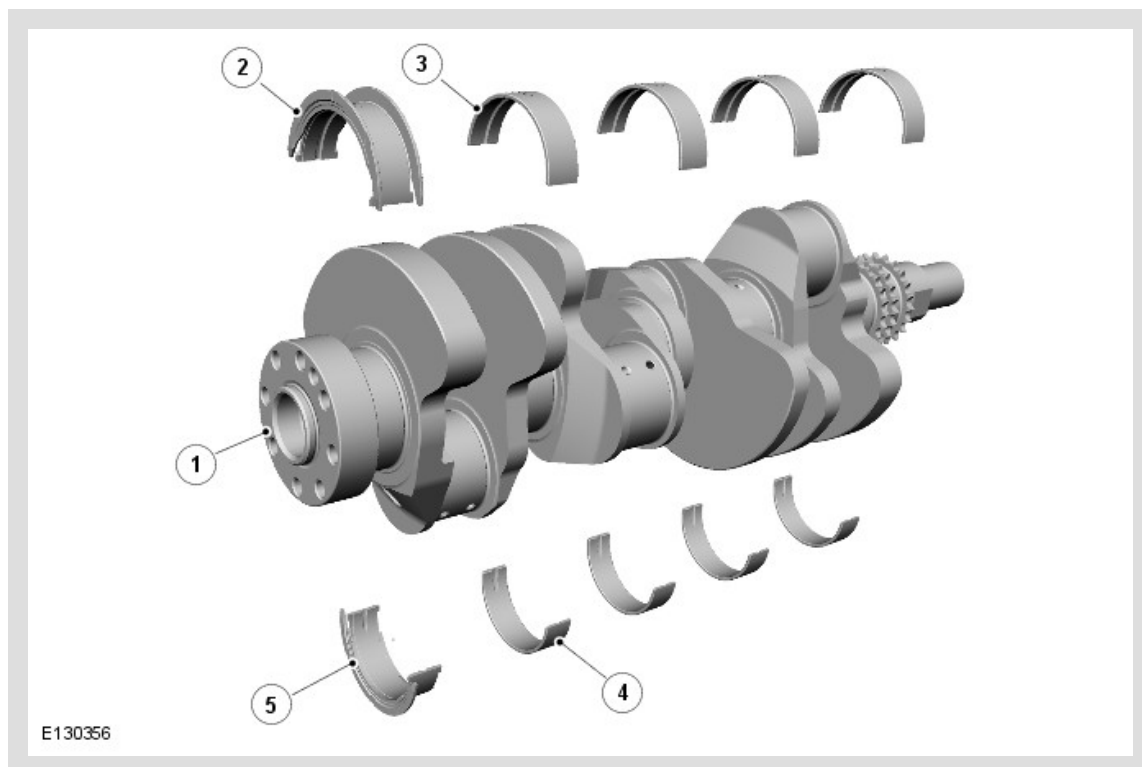
## CRANKSHAFT AND OIL PAN COMPONENTS



| ITEM | DESCRIPTION                                                                           |
|------|---------------------------------------------------------------------------------------|
| 1    | Coolant pump                                                                          |
| 2    | Cylinder block                                                                        |
| 3    | Crankshaft rear oil seal                                                              |
| 4    | Drive plate                                                                           |
| 5    | Main bearings (10 off, 5 upper and 5 lower. Number 5 bearings include thrust washers) |
| 6    | Main bearing caps (5 off)                                                             |
| 7    | Main bearing cap cross bolts (10 off)                                                 |

|    |                                                           |
|----|-----------------------------------------------------------|
| 8  | Main bearing cap bolts (20 off)                           |
| 9  | Oil baffle plate                                          |
| 10 | Oil scavenge pipe                                         |
| 11 | Oil pick-up and strainer                                  |
| 12 | Oil pan gasket                                            |
| 13 | Oil level and temperature sensor                          |
| 14 | Oil pan                                                   |
| 15 | Oil drain plug                                            |
| 16 | Oil pan bolt (22 off)                                     |
| 17 | Bolt (16 off, 2 per piston)                               |
| 18 | Connecting rod cap                                        |
| 19 | Connecting rod big end bearing (2 off per connecting rod) |
| 20 | Connecting rod                                            |
| 21 | Connecting rod small end bearing                          |
| 22 | Piston                                                    |
| 23 | Piston pin retainer (16 off, 2 per piston)                |
| 24 | Piston pin                                                |
| 25 | Oil control ring                                          |
| 26 | Lower compression ring                                    |
| 27 | Upper compression ring                                    |
| 28 | Bolt                                                      |
| 29 | Crankshaft damper pulley                                  |
| 30 | Oil pump                                                  |
| 31 | Crankshaft                                                |
| 32 | Diamond washer                                            |

## CRANKSHAFT AND MAIN BEARINGS



| ITEM | DESCRIPTION                                        |
|------|----------------------------------------------------|
| 1    | Crankshaft                                         |
| 2    | Rear main bearing shell - upper with thrust washer |
| 3    | Main bearing shell - upper (4 off)                 |
| 4    | Main bearing shell - lower (4 off)                 |
| 5    | Rear main bearing shell - lower with thrust washer |

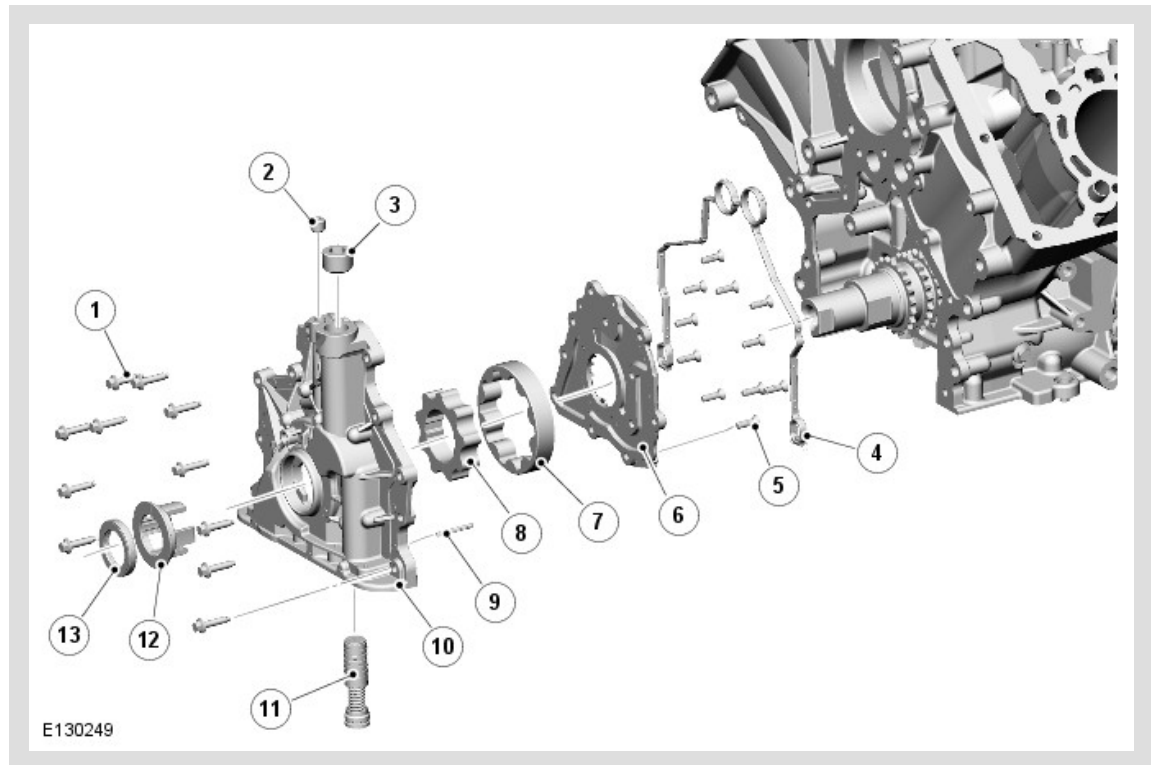
The crankshaft is installed in five main bearings. The main bearing caps are double and cross-bolted, which adds to the strength and rigidity of the engine block.

The crankshaft drive pulley is not keyed onto the crankshaft; the crankshaft and the pulley hub have a corresponding flat which provides positive location. The pulley is secured to the crankshaft with a bolt and diamond washer.

The main bearings are selective bearings. An oil groove in the top half of each bearing transfers oil into the crankshaft for lubrication of the connecting rod bearings. The

upper and lower shells of bearing number 5 contain integral thrust washers, which limits the crankshaft end float.

#### CRANKSHAFT FRONT OIL SEAL AND OIL PUMP



| ITEM | DESCRIPTION                                                                    |
|------|--------------------------------------------------------------------------------|
| 1    | Bolt (11 off)                                                                  |
| 2    | Clean oil gallery plug                                                         |
| 3    | Outlet gallery plug                                                            |
| 4    | Gasket                                                                         |
| 5    | Cover screw (11 off)                                                           |
| 6    | Backplate                                                                      |
| 7    | Oil pump outer rotor                                                           |
| 8    | Oil pump inner rotor                                                           |
| 9    | Roll pin                                                                       |
| 10   | Oil pump housing                                                               |
| 11   | Relief valve                                                                   |
| 12   | Oil pump insert (fitting tool only - discarded after installation of oil pump) |
| 13   | Crankshaft front oil seal                                                      |

The oil pump draws engine oil out via a strainer in the oil pan, through the oil filter, galleries, and to the engine bearings, pistons and turbochargers.

The oil pump is located at the front of the engine in an oil pump housing which is secured to the cylinder block with 11 bolts. The pump is an eccentric rotor type pump which is located in the rear of the housing and sealed with a backplate and screws.

The pump is driven by a direct drive off the crankshaft.

The oil pump housing also contains a relief valve assembly which is secured in the housing with a roll pin. The oil pump supplies a greater amount of oil than is normally required for adequate lubrication. The oil pressure relief valve is incorporated in the pump oil circuit to limit maximum oil pressure to 4.5 bar (65.25 lbf/in<sup>2</sup>).

The pressure relief valve is a spring-loaded bypass valve which comprises a small piston and spring located in a bore in the oil pump housing. Under normal pressure conditions, the spring holds the relief valve closed and all the oil from the oil pump flows into the oil galleries and to the bearings. Under abnormally high oil pressure conditions (when the oil is cold for example), the pressure relief valve opens. Oil pressure pushes the small piston back in its bore in the housing by overcoming spring tension. This allows some of the pressurized oil to bypass the main oil galleries and return back into the oil pan. Most of the oil still flows to the bearings and a preset oil pressure is maintained.

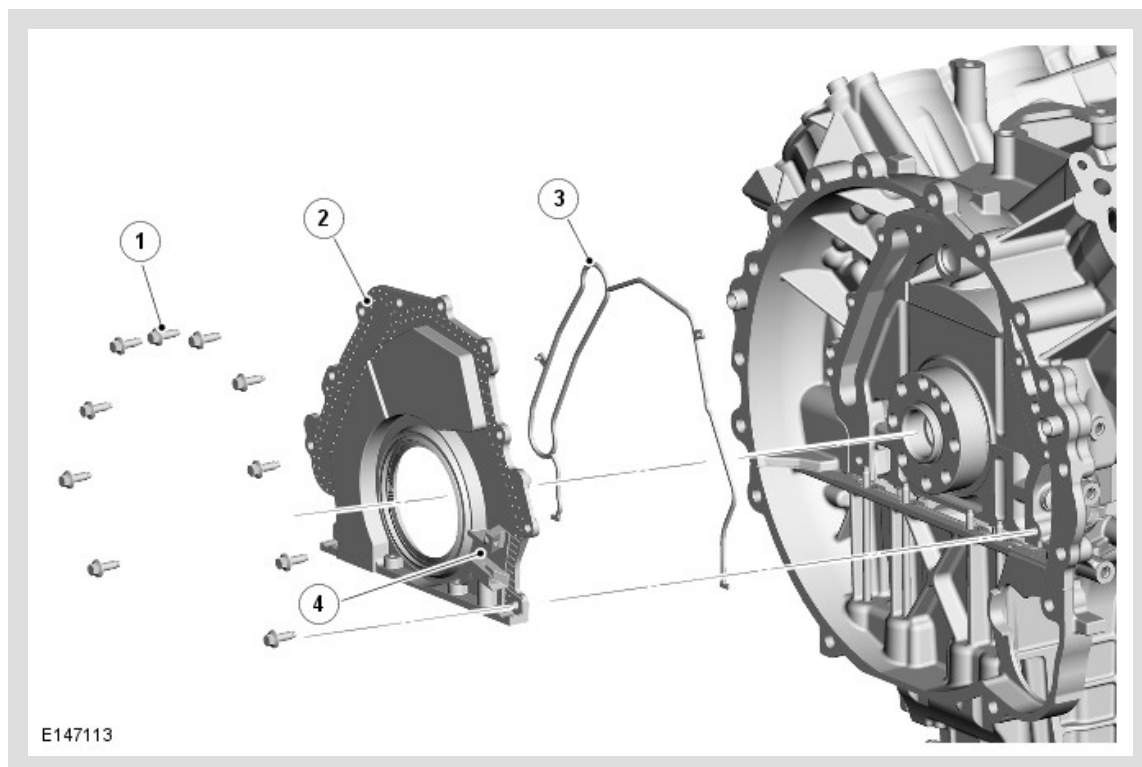
The crankshaft front oil seal is a press fit into the oil pump housing and requires a special tool for fitment.



**NOTE:**

The seal must be inserted such that the front face is 1 mm underflush with the machined front face of the oil pump. It must not be pushed all the way into the bore, as this will block the seal drains.

## CRANKSHAFT REAR OIL SEAL

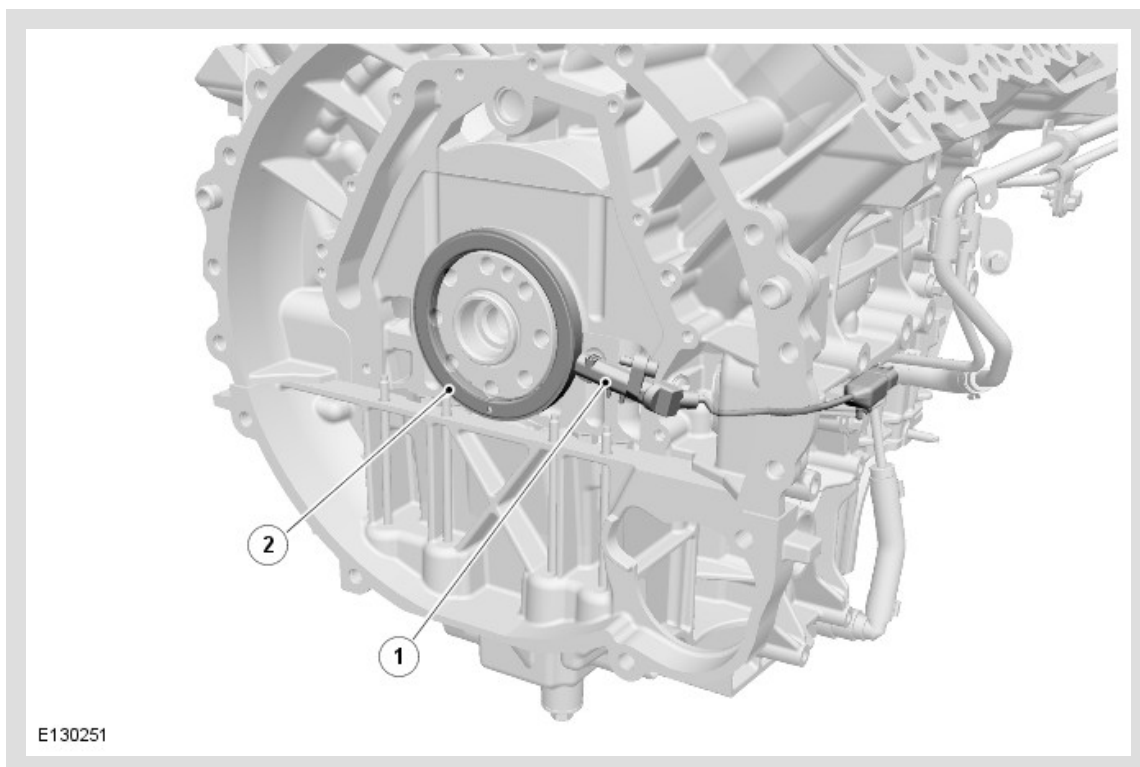


| ITEM | DESCRIPTION                         |
|------|-------------------------------------|
| 1    | Bolt (10 off)                       |
| 2    | Crankshaft rear oil seal            |
| 3    | Gasket                              |
| 4    | Crankshaft position sensor location |

The crankshaft rear oil seal consists of a lipped oil seal pressed into a retainer. The retainer is attached to the rear of the crankcase with 10 bolts.

The retainer also provides the location for the CKP sensor.

#### RELUCTOR RING



| ITEM | DESCRIPTION                              |
|------|------------------------------------------|
| 1    | Crankshaft position sensor reluctor ring |
| 2    | Reluctor ring                            |

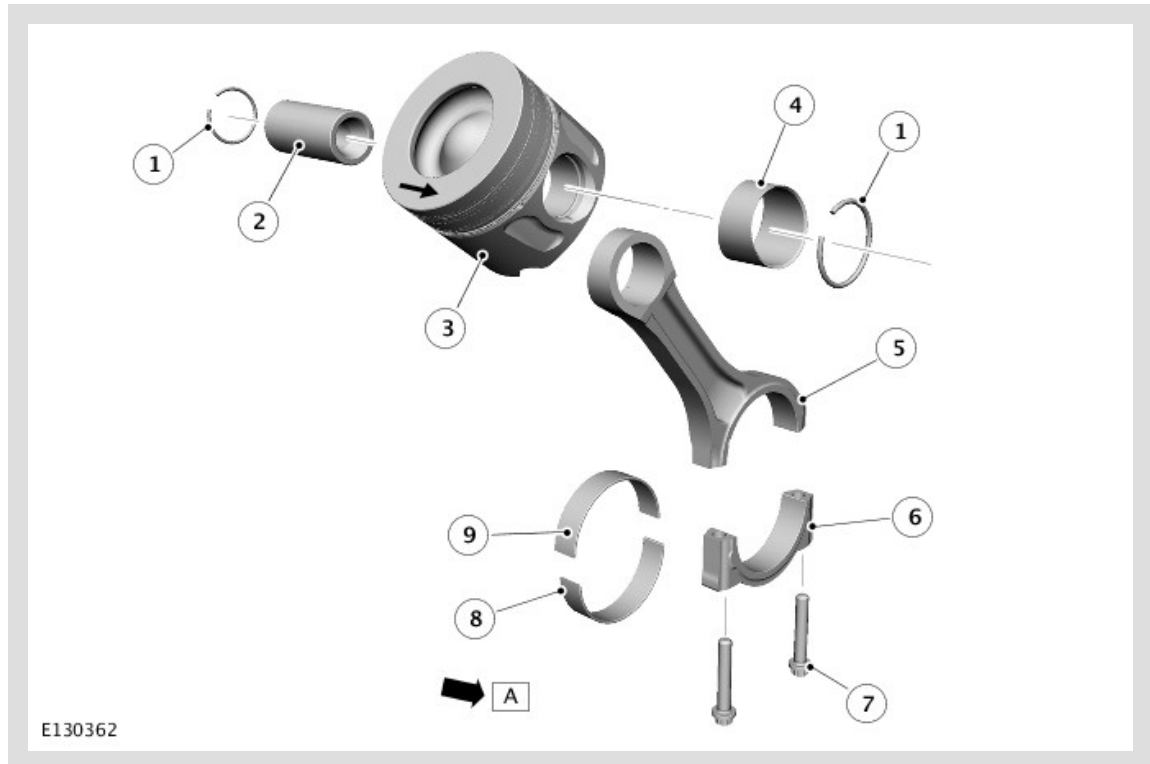
The reluctor ring is located on the rear of the crankshaft. It is pressed onto the crankshaft using a special tool, which precisely aligns the reluctor ring for crankshaft position and timing. The reluctor ring consists of 60 equally spaced points, 58 of which contain a magnet. The 2 missing spaces are used for ECM crankshaft position reference and synchronisation. The magnets cannot be seen on the trigger-wheel; therefore, it can only be positioned using the special tool.



#### NOTE:

If the reluctor ring is removed for any reason, a new reluctor ring must be fitted. Do not reuse the old reluctor ring.

## CONNECTING RODS AND PISTONS



| ITEM | DESCRIPTION                      |
|------|----------------------------------|
| A    | Front of engine                  |
| 1    | Piston pin retainers             |
| 2    | Piston pin                       |
| 3    | Piston                           |
| 4    | Connecting rod small end bearing |
| 5    | Connecting rod                   |
| 6    | Connecting rod bearing cap       |
| 7    | Bolt (2 off)                     |
| 8    | Connecting rod bearing (lower)   |
| 9    | Connecting rod bearing (upper)   |

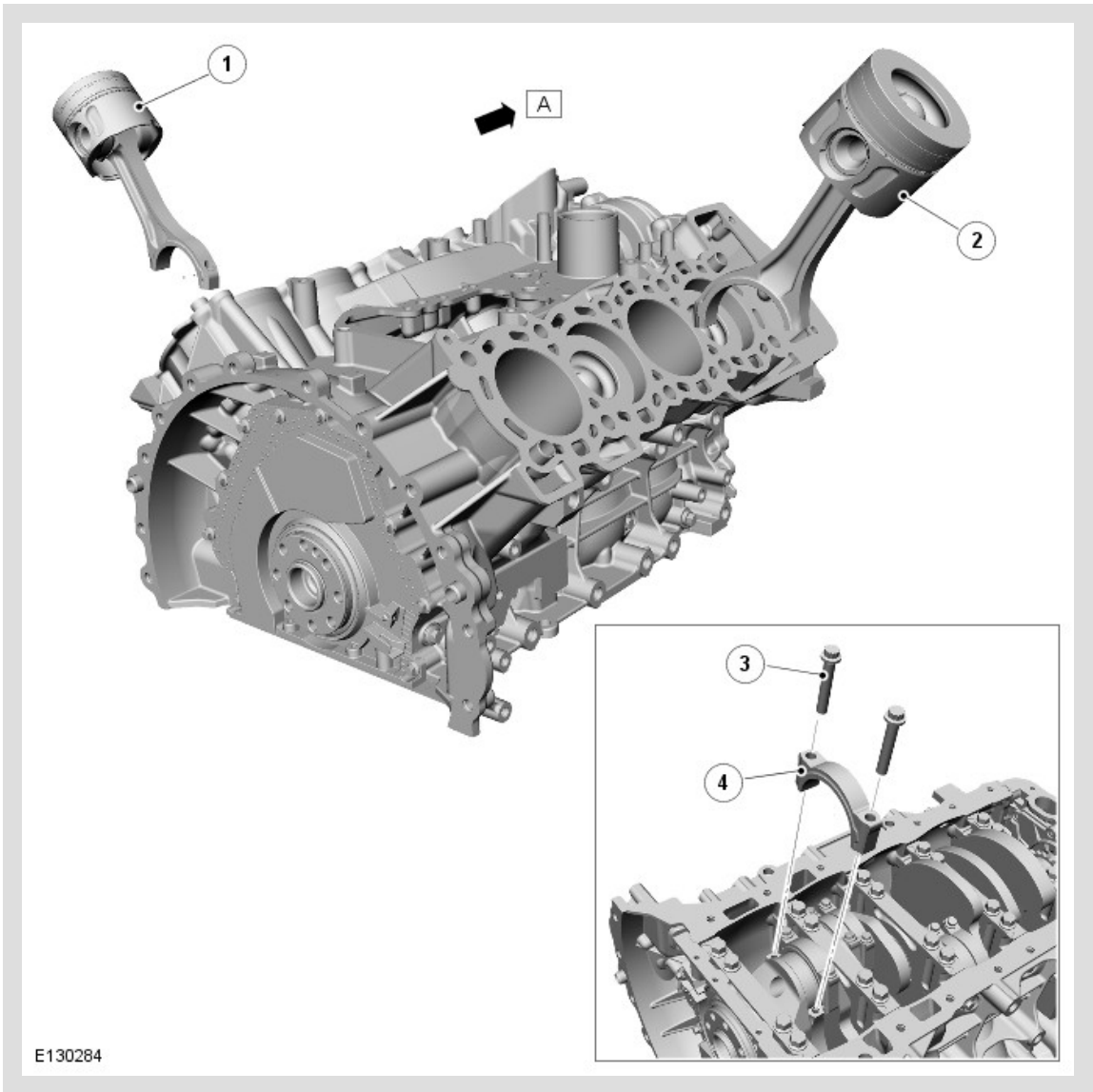
The connecting rods have fracture-split bearing caps. Fracturing the opposing sides of the connecting rod at the bearing horizontal centerline, produces the bearing caps. As

well as being easier to manufacture, when reassembled the fractured surfaces interlock to form a strong seamless joint. The cylinder position is etched on adjoining sides of the joint to identify matching connecting rods and bearing caps. The selective connecting rod bearings are split plain bearings.



**NOTE:**

The connecting rods are not selective.



| ITEM | DESCRIPTION                                       |
|------|---------------------------------------------------|
| A    | Front of engine                                   |
| 1    | Piston and connecting rod assembly, cylinders 5-8 |
| 2    | Piston and connecting rod assembly, cylinders 1-4 |

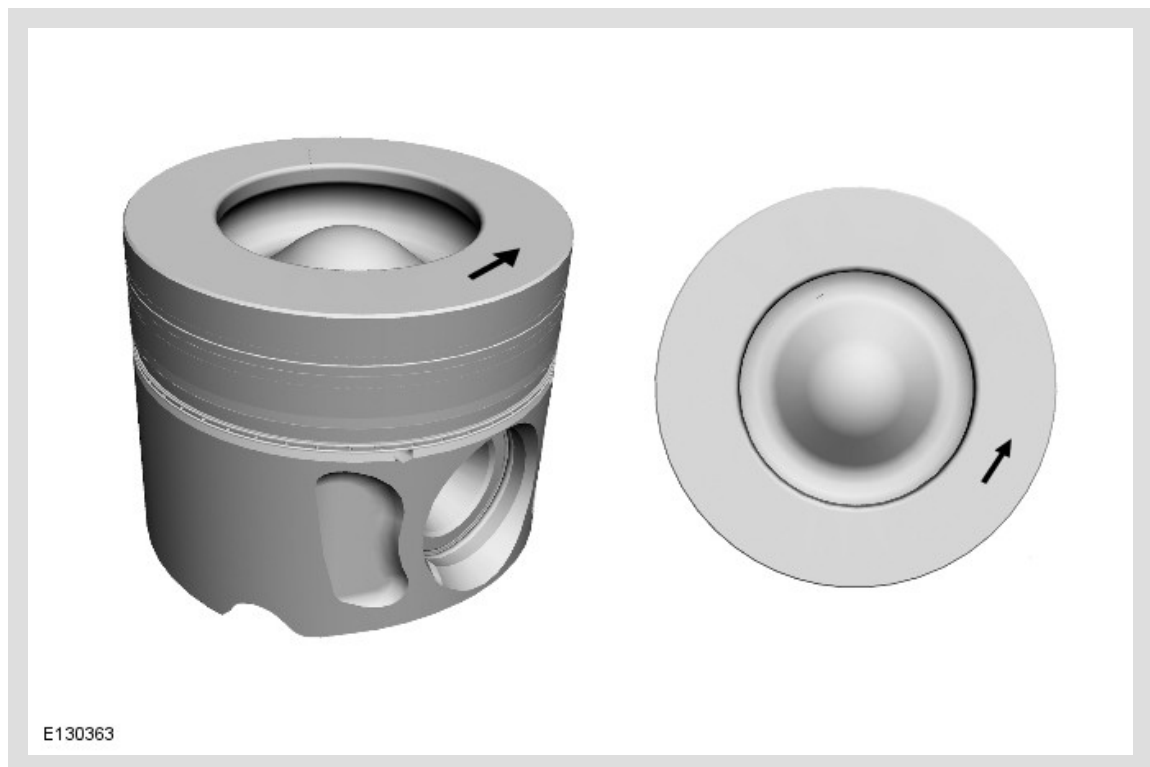
|   |                                    |
|---|------------------------------------|
| 3 | Bolt (16 off)                      |
| 4 | Connecting rod bearing cap (8 off) |

When installing a connecting rod, ensure the back of the connecting rod faces the center of the 'vee'.

The pistons are fitted with 2 compression rings and an oil control ring. The piston crown incorporates a pronounced bowl; this forms the combustion chamber, which promotes swirl and turbulence necessary for good combustion and improved emissions. In addition, the piston skirt has a molybdenum-coated surface, which counteracts scoring of the cylinder bore and piston.

The piston also incorporates a double wave gallery within the underside of the piston crown to enhance piston cooling. The pistons are supplied oil by means of spray jets located in the cylinder block oil gallery. These jets ensure optimum piston cooling to counteract the high temperatures generated by the combustion process.

Each piston is installed on a piston pin, which is located in a bushing in the connecting rod.



When installing pistons ensure the arrows on the piston crowns all point to the front of the engine.



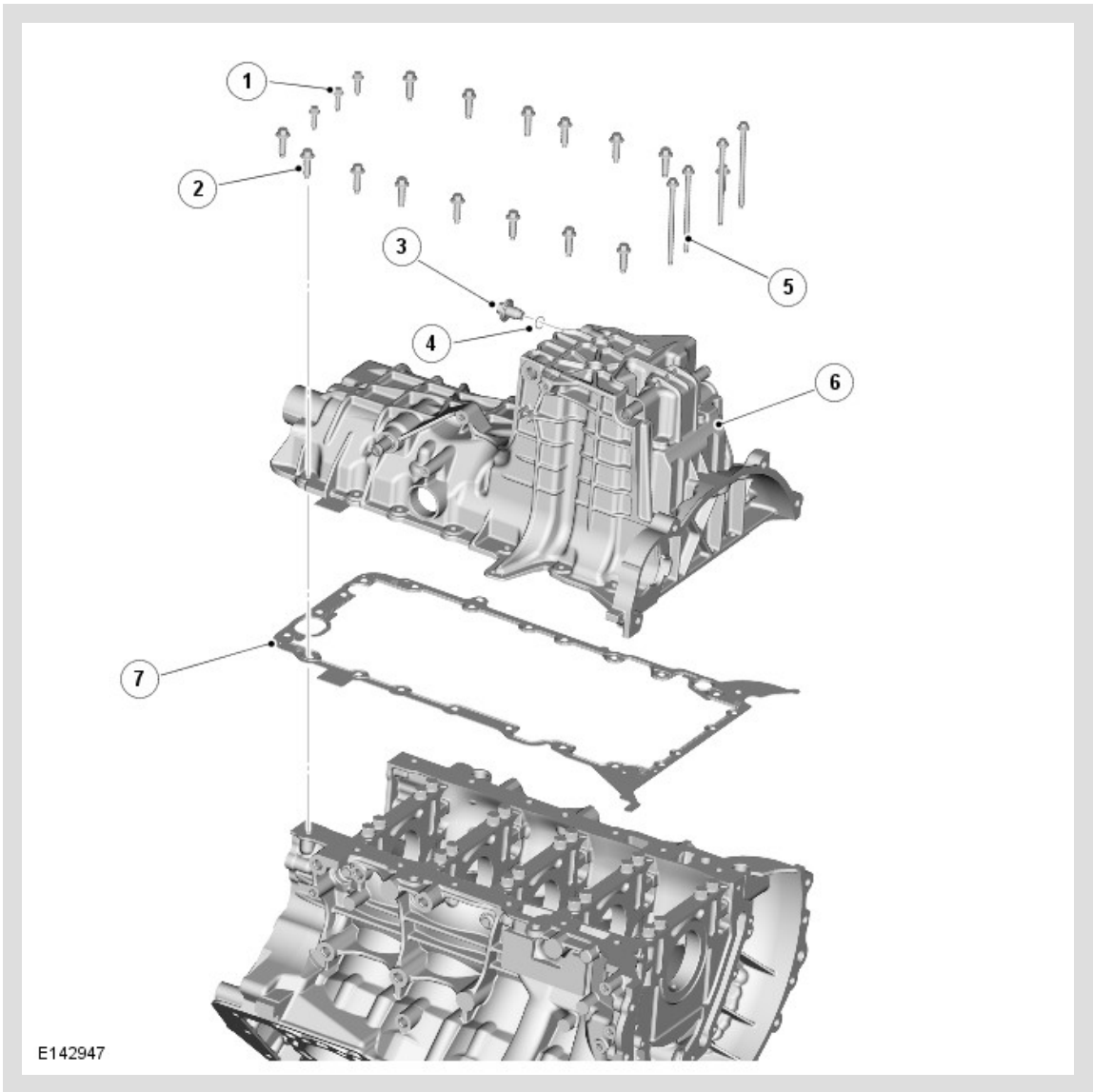
#### NOTE:

All pistons are common single grade/single part number.

The piston top ring is a taper type and is fitted with the taper to the top of the piston. All rings marked 'top' are assembled with 'top' uppermost. All rings must be spaced evenly around the piston before installing, with the ring gaps at 120° to each other. The circumference gap of the double beveled oil control ring must be opposite the spiral control joint.

The upper and lower compression rings and the oil control ring are identified by a paint code on each ring. The upper compression ring has a red paint mark, the lower compression ring has a yellow paint mark and the oil control ring has a light blue paint mark.

OIL PAN

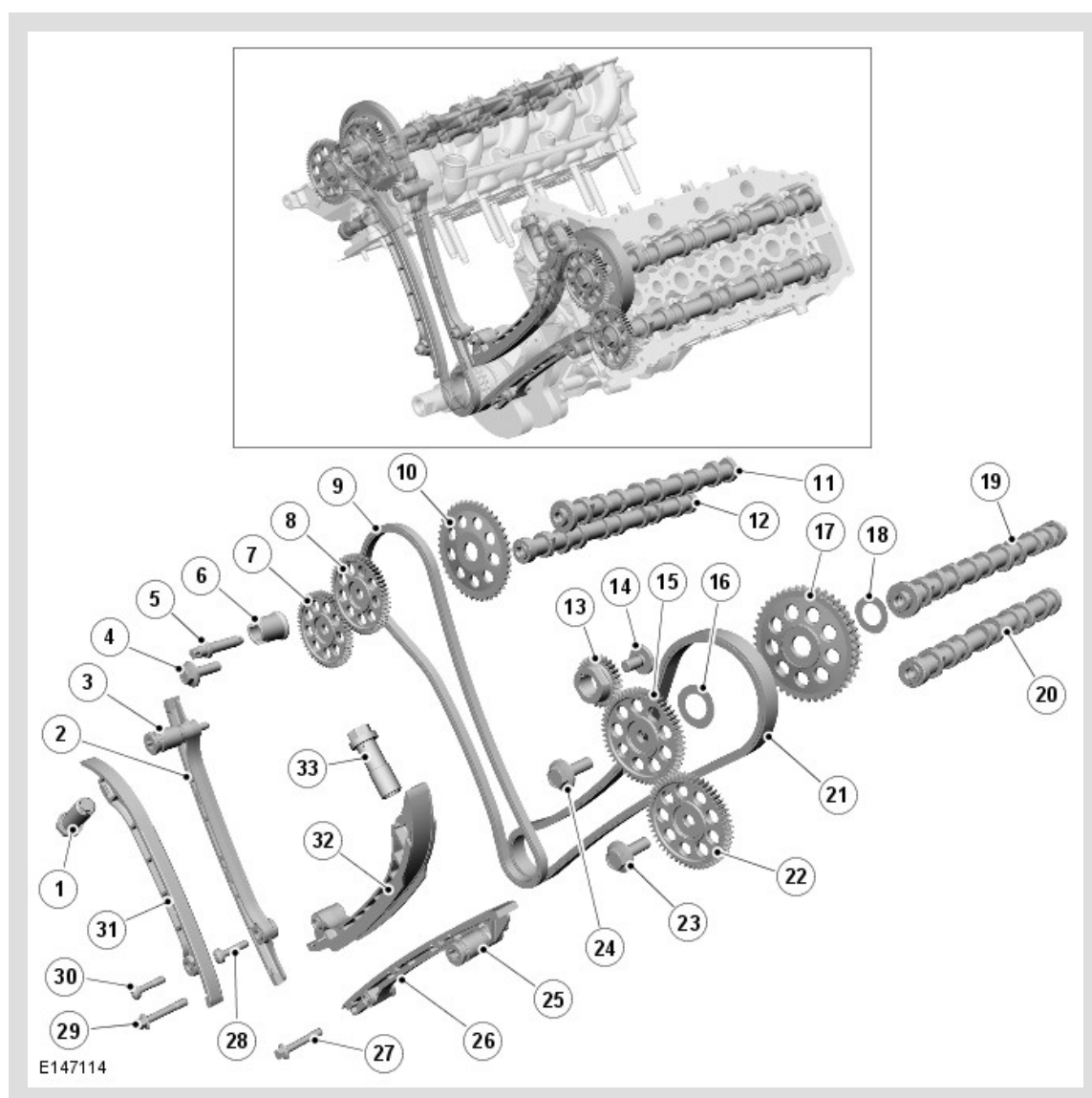


| ITEM | DESCRIPTION   |
|------|---------------|
| 1    | Bolt (3 off)  |
| 2    | Bolt (14 off) |

|   |                |
|---|----------------|
| 3 | Oil drain plug |
| 4 | Seal           |
| 5 | Bolt (4 off)   |
| 6 | Oil pan        |
| 7 | Oil pan gasket |

The oil pan is heavily ribbed for strength and stiffness. An oil drain plug is located on the underside of the oil pan. A gasket seals the joint between the oil pan and the cylinder block.

## CAMSHAFT TIMING COMPONENTS



| ITEM | DESCRIPTION                    |
|------|--------------------------------|
| 1    | Right camshaft chain tensioner |
|      |                                |

|    |                                    |
|----|------------------------------------|
| 2  | Right camshaft chain guide         |
| 3  | Bolt                               |
| 4  | Bolt                               |
| 5  | Bolt - camshaft drive gear adaptor |
| 6  | Vacuum and oil scavenge pump drive |
| 7  | Right exhaust camshaft gear        |
| 8  | Right intake camshaft gear         |
| 9  | Timing chain                       |
| 10 | Right intake camshaft sprocket     |
| 11 | Right intake camshaft              |
| 12 | Right exhaust camshaft             |
| 13 | High pressure fuel pump drive gear |
| 14 | Bolt                               |
| 15 | Left intake camshaft gear          |
| 16 | Diamond washer                     |
| 17 | Left intake camshaft sprocket      |
| 18 | Diamond washer                     |
| 19 | Left intake camshaft               |
| 20 | Left exhaust camshaft              |
| 21 | Timing chain                       |
| 22 | Left exhaust camshaft gear         |
| 23 | Bolt                               |
| 24 | Bolt                               |
| 25 | Bolt                               |
| 26 | Left camshaft chain guide          |
| 27 | Bolt                               |
| 28 | Bolt                               |
| 29 | Bolt                               |
| 30 | Bolt                               |
| 31 | Right camshaft chain tensioner arm |
| 32 | Left camshaft chain tensioner arm  |
| 33 | Left camshaft chain tensioner      |

Drive is provided by a simplex bush timing chain for the right camshafts and a duplex bush timing chain for the left camshafts. The two timing chains are driven from the crankshaft to the intake camshaft sprocket of each cylinder head. Each exhaust camshaft is driven by a gear arrangement from the intake camshaft.

On the left cylinder head, the gear arrangement also drives the high pressure fuel pump. To ensure there is no slippage in the drive from the intake camshaft sprocket, due to the load of the fuel pump, diamond washers are installed each side of the sprocket. The diamond washers increase the friction between the sprocket and the adjacent components.



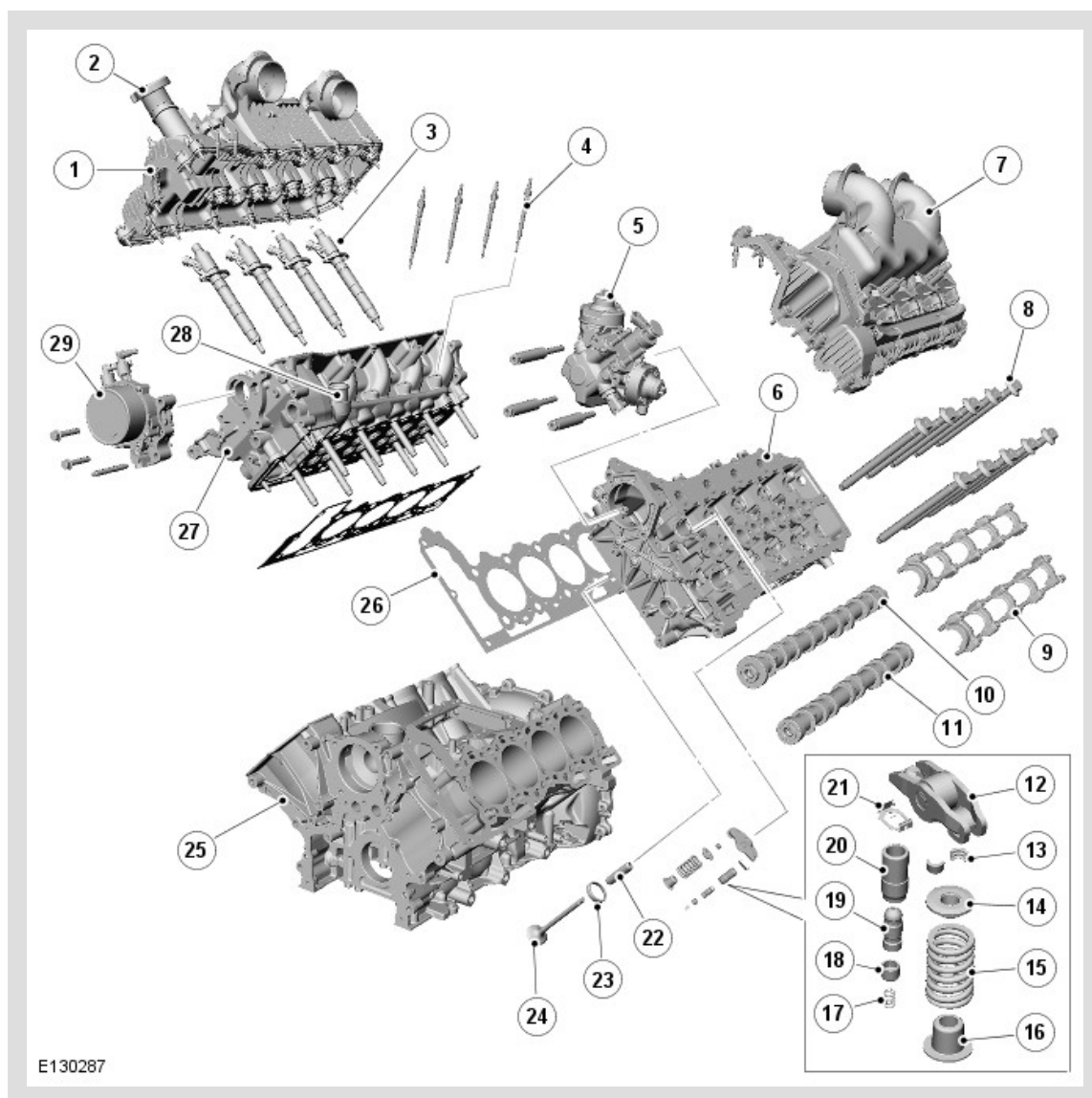
**NOTE:**

The diamond washers can become embedded in the sprocket, and difficult to separate when the timing gear is disassembled.

The non-driven side of each timing chain is contained between fixed and moveable camshaft chain guides. The moveable guides are attached to the front of the cylinder block using pivot bolts. Hydraulic tensioners keep the moveable guides in contact with the timing chains, to control timing chain flutter.

The timing chains are oil splash lubricated. Oil spray is directed onto the chains from several oil supply ports in the front of the cylinder block and cylinder heads.

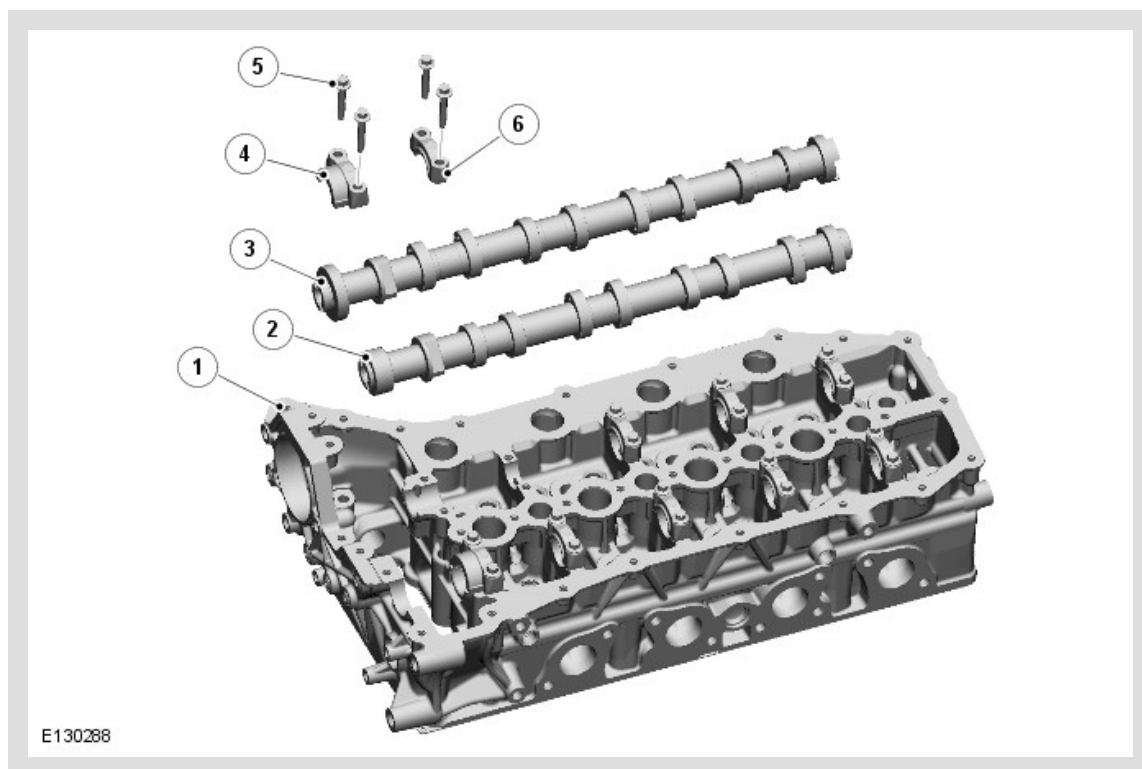
## **CYLINDER HEAD COMPONENTS**



| ITEM | DESCRIPTION                                              |
|------|----------------------------------------------------------|
| 1    | Right camshaft cover                                     |
| 2    | Oil filler cap                                           |
| 3    | Fuel injectors (4 off per cylinder head)                 |
| 4    | Glow plugs (4 off per cylinder head)                     |
| 5    | High pressure fuel pump                                  |
| 6    | Left cylinder head                                       |
| 7    | Left camshaft cover                                      |
| 8    | Cylinder head bolts (10 off per cylinder head)           |
| 9    | Cam caps and bolts assemblies (10 off per cylinder head) |
| 10   | Left intake camshaft                                     |
| 11   | Left exhaust camshaft                                    |
| 12   | Rocker arm (16 off per cylinder head)                    |

|    |                                                            |
|----|------------------------------------------------------------|
| 13 | Valve spring retaining collet (32 off per cylinder head)   |
| 14 | Valve spring retainer (16 off per cylinder head)           |
| 15 | Valve spring (16 off per cylinder head)                    |
| 16 | Valve spring seat and stem seal (16 off per cylinder head) |
| 17 | Rocker arm - Return spring (16 off per cylinder head)      |
| 18 | Rocker arm - Lower piston (16 off per cylinder head)       |
| 19 | Rocker arm - Upper piston (16 off per cylinder head)       |
| 20 | Rocker arm - housing (16 off per cylinder head)            |
| 21 | Clip (16 off per cylinder head)                            |
| 22 | Valve guide (16 off per cylinder head)                     |
| 23 | Valve seat (16 off per cylinder head)                      |
| 24 | Valve (8 off exhaust, 8 off intake per cylinder head)      |
| 25 | Cylinder block                                             |
| 26 | Cylinder head gasket (1 off per cylinder head)             |
| 27 | Right cylinder head                                        |
| 28 | Engine coolant supply connection                           |
| 29 | Vacuum and oil scavenge pump                               |

## CYLINDER HEADS



| ITEM | DESCRIPTION                                      |
|------|--------------------------------------------------|
| 1    | Cylinder head (left shown, right similar)        |
| 2    | Exhaust camshaft                                 |
| 3    | Intake camshaft                                  |
| 4    | Camshaft bearing cap front (1 off per camshaft)  |
| 5    | Camshaft bearing cap bolts (10 off per camshaft) |
| 6    | Camshaft bearing cap (4 off per camshaft)        |

The cylinder heads are unique to each cylinder. Ten deep-seated bolts help reduce distortion and secure each cylinder head to the cylinder block. The cylinder head bolts are located beneath the camshafts, 5 under the intake camshaft and 5 under the exhaust camshaft. Two hollow dowels align each cylinder head with the cylinder block.



#### NOTES:

- The cylinder head bolts are not accessible with the camshafts fitted.
- The cylinder head cannot be reworked.

Each cylinder head has 4 ports machined at each cylinder location, 2 exhaust ports and 2 intake ports. One of the intake ports is helical and functions as a swirl port, the other is arranged laterally as a tangential port and functions as a charge port.

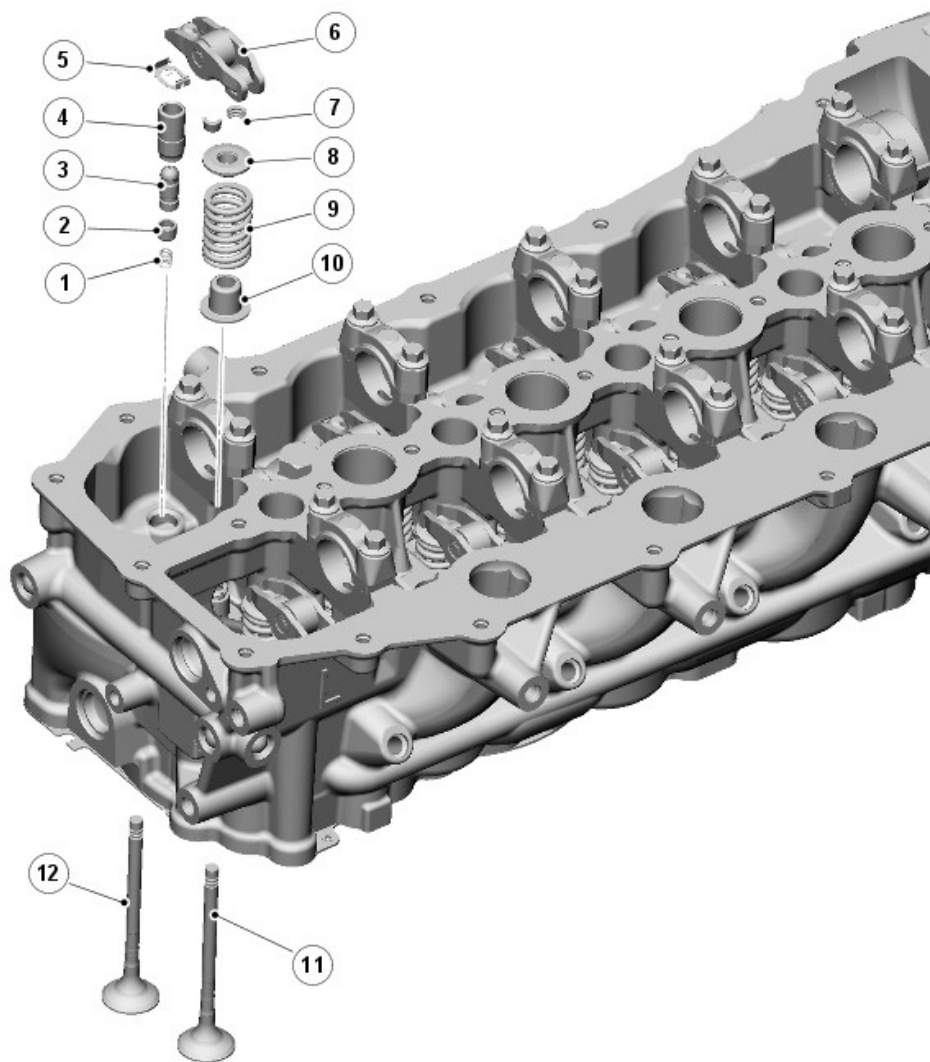
Each cylinder head houses a manual chain tensioner that operates on the adjustable guide at the slack side of each of the timing chains. Pressurized oil for the adjuster is supplied through the back of the unit from an oil supply port in each cylinder head.

## CAMSHAFTS

There is 1 exhaust camshaft and 1 intake camshaft per cylinder head. Each of the overhead camshafts are located in 5 bearings and maintained in position by 5 bearing caps. Each of the bearing caps are fixed to the cylinder head by 2 bolts. Location letters and numbers are etched on the caps for each cylinder head, A to L for the left head and 0 to 9 for the right head. The camshafts are of a hollow tube construction, with pressed on lobes. The camshafts are driven from the crankshaft via a gear arrangement and a simplex chain for the right cylinder head camshaft and a duplex chain for the left cylinder head camshafts.

Each camshaft has 8 machined lobes for operating the intake and exhaust valves through hydraulic lash adjusters and roller-type finger levers. The front of the right cylinder head intake camshaft provides a drive connection for the vacuum and oil scavenge pump, via a drive gear. The front of the left cylinder head intake camshaft provides a drive connection for the high-pressure fuel pump, also via a drive gear.

## VALVES AND LASH ADJUSTERS



E130289

| ITEM | DESCRIPTION                                    |
|------|------------------------------------------------|
| 1    | Return spring (16 off per cylinder head)       |
| 2    | Lower piston (16 off per cylinder head)        |
| 3    | Upper piston (16 off per cylinder head)        |
| 4    | Housing (16 off per cylinder head)             |
| 5    | Clip (16 off per cylinder head)                |
| 6    | Rocker arm assembly (16 off per cylinder head) |
| 7    | Valve spring collet (32 off per cylinder head) |

|    |                                                  |
|----|--------------------------------------------------|
| 8  | Valve spring retainer (16 off per cylinder head) |
| 9  | Valve spring (16 off per cylinder head)          |
| 10 | Valve stem seal (16 off per cylinder head)       |
| 11 | Intake valve (8 off per cylinder head)           |
| 12 | Exhaust valve (8 off per cylinder head)          |

Two 27.5 mm intake valves and 2 25.0 mm exhaust valves serve each cylinder. Each valve is of a ground, solid one-piece cylinder head construction.

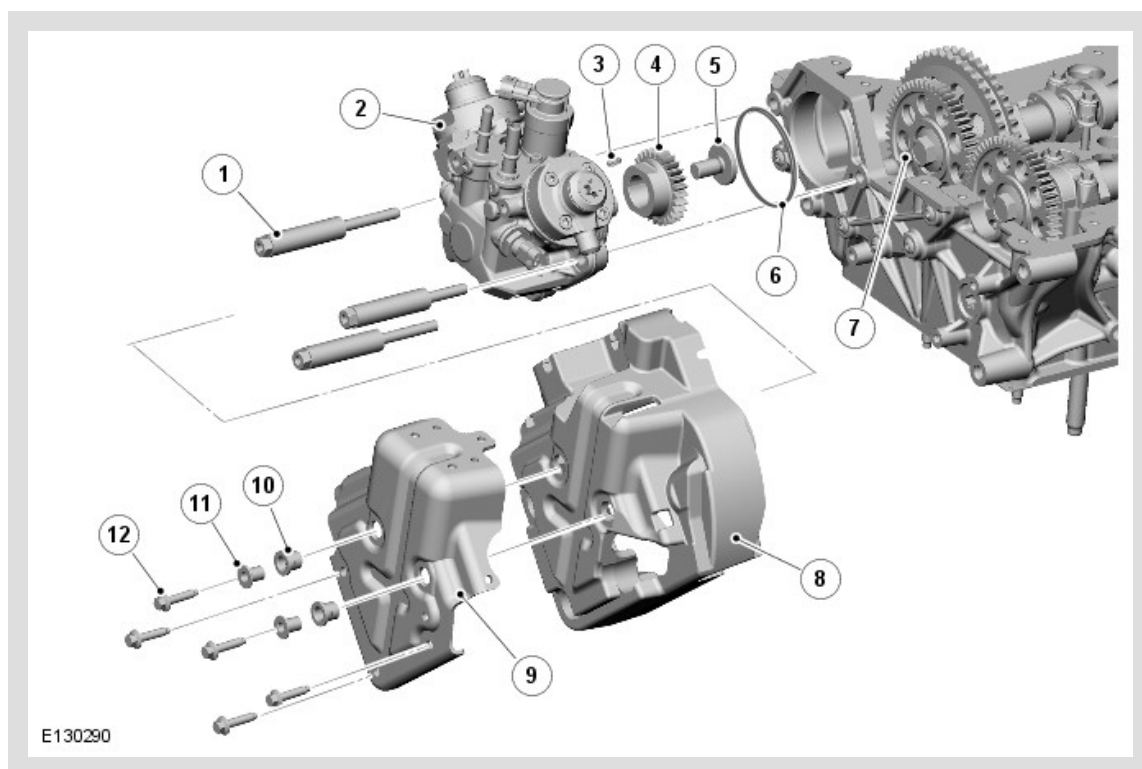
The valve springs are of the parallel single-coil type. The bottom end of each spring rests on the flange of a spring retainer, which has an integral valve stem seal. The top end of the spring is held in place by a spring retainer, which is held in position at the top end of the valve stem by split taper collets. The taper collets have grooves on the internal bore that locate to grooves ground into the upper stems of the valves.

The valve spring seats and guides are an interference fit in the cylinder head.

| VALVE                | POSITION                              |
|----------------------|---------------------------------------|
| Intake valve opens   | 8.5 degrees Before Top Dead Center    |
| Intake valve closes  | 35.5 degrees After Bottom Dead Center |
| Exhaust valve opens  | 64 degrees Before Bottom Dead Center  |
| Exhaust valve closes | 12 degrees After Top Dead Center      |

The valves are operated through roller-type rocker arms and hydraulic lash adjusters, actuated by the camshaft lobes. When the camshaft lobe presses down on the top of a rocker arm roller mechanism, the respective valve is forced down, opening the effected intake or exhaust port. The use of this type of actuation method helps reduce friction in the valve timing mechanism.

#### HIGH PRESSURE FUEL PUMP



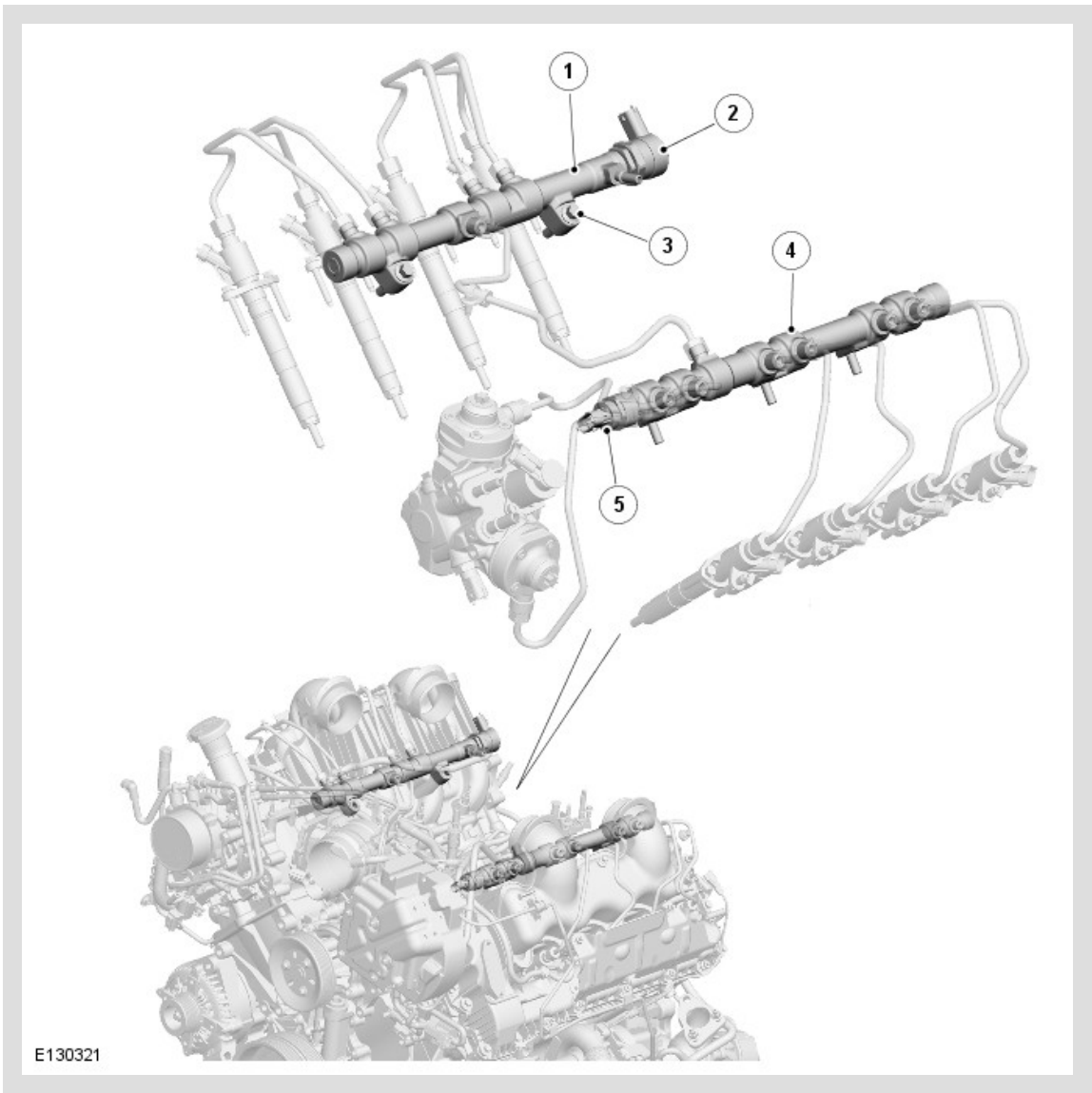
| ITEM | DESCRIPTION                |
|------|----------------------------|
| 1    | Bolt (3 off)               |
| 2    | High-pressure fuel pump    |
| 3    | Woodruff key               |
| 4    | Fuel pump drive gear       |
| 5    | Fuel pump gear flange bolt |
| 6    | O-ring seal                |
| 7    | Intake camshaft drive gear |
| 8    | Sound absorbing cover      |
| 9    | Shield                     |
| 10   | Mounting sleeve (2 off)    |
| 11   | Sleeve (2 off)             |
| 12   | Bolt (5 off)               |

The high-pressure fuel pump supplies the common rails with fuel and is fixed to the front of the left cylinder head with 3 special bolts.

The pump is a 2-radial piston type and is driven by the intake camshaft drive gear of the left cylinder head via the fuel pump drive gear. The fuel pump drive gear is located on the pump input shaft with a woodruff key and the gear is secured using a flanged bolt.

An O-ring seal is used to seal the pump body to the fuel pump mounting bracket which is attached to the front of the left cylinder head. A sound absorbing cover is attached to the front of the pump body and is secured with a shield retained with 5 bolts. For additional information, refer to: [Fuel Charging and Controls](#) (303-04F Fuel Charging and Controls - TDV8 4.4L Diesel, Description and Operation).

HIGH PRESSURE FUEL RAILS



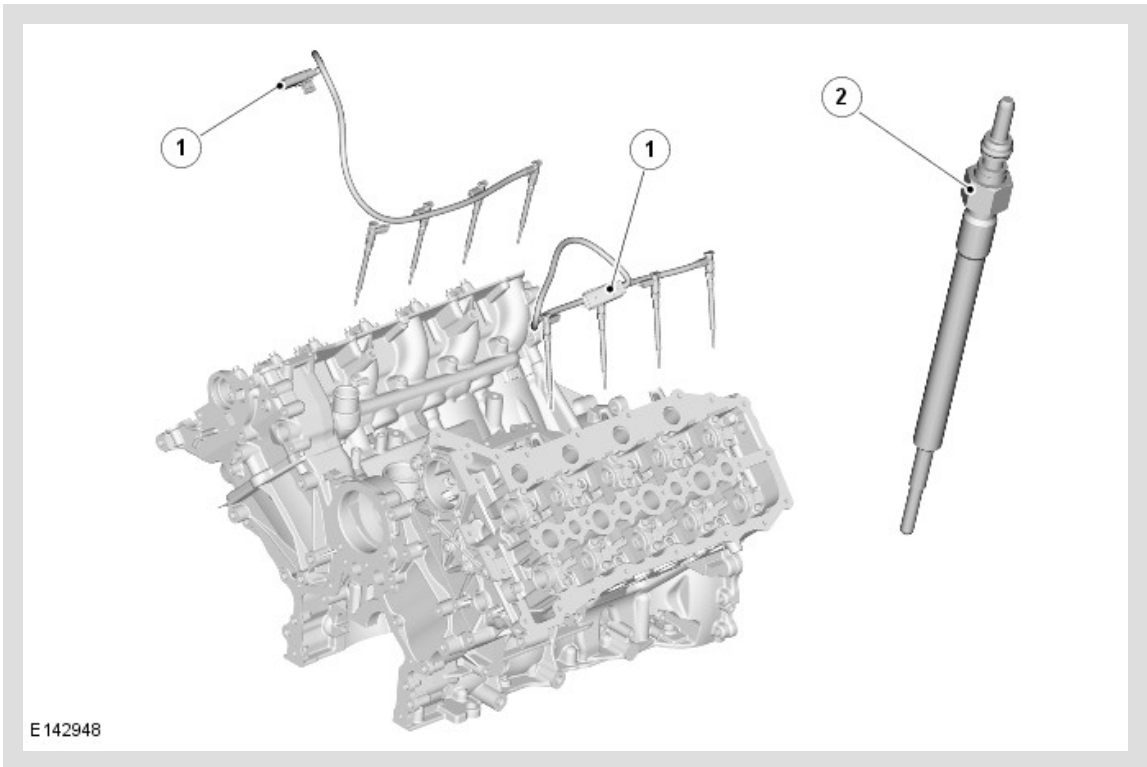
| ITEM | DESCRIPTION                   |
|------|-------------------------------|
| 1    | Right high pressure fuel rail |

|   |                              |
|---|------------------------------|
| 2 | Fuel pressure control valve  |
| 3 | Bolt (4 off, 2 per rail)     |
| 4 | Left high pressure fuel rail |
| 5 | Fuel rail pressure sensor    |

The fuel rails store the fuel at high pressure and prevent pressure fluctuations in the high-pressure system. A fuel rail pressure sensor is fitted into a threaded bore in the forward end of the left fuel rail. A fuel pressure control valve is located in the rearward facing end of the right fuel rail.

For additional information, refer to: [Fuel Charging and Controls](#) (303-04F Fuel Charging and Controls - TDV8 4.4L Diesel, Description and Operation).

### GLOW PLUGS



| ITEM | DESCRIPTION                         |
|------|-------------------------------------|
| 1    | Glow plug harness connector         |
| 2    | Glow plug (4 off per cylinder head) |

The glow plugs are arranged centrally on the intake side of the cylinder head, between the 2 intake ports of each cylinder.

For additional information, refer to: [Glow Plug System](#) (303-07D Glow Plug System - TDV8 4.4L Diesel, Description and Operation).



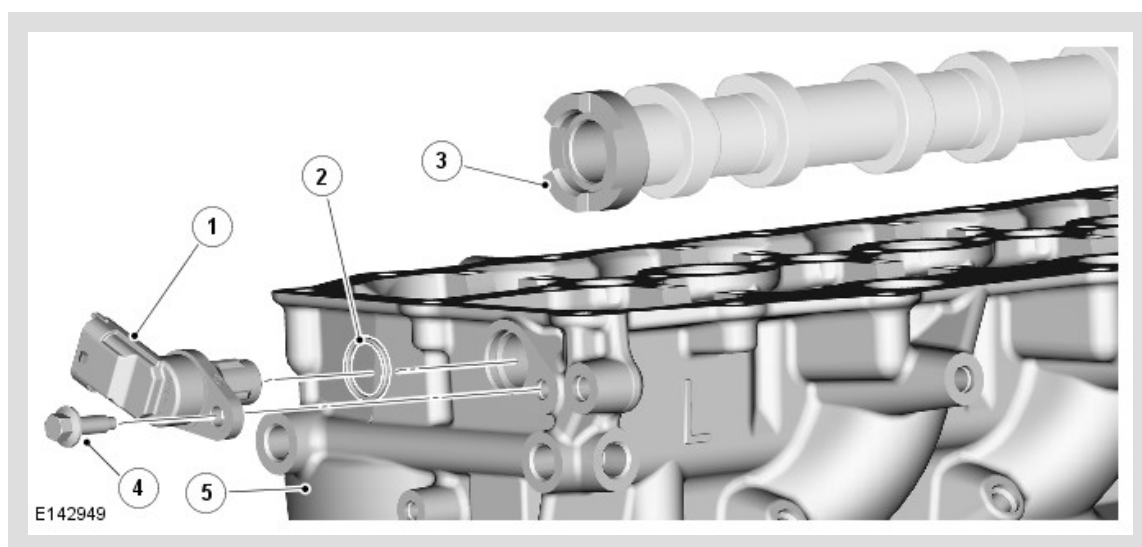
### CAUTION:

The glow plugs are sensitive to damage due to the material properties of the ceramic heater pin. The ceramic glow plugs are sensitive to shock and bending and for this reason a dropped glow plug **MUST NOT** be used, even if there is no visible damage.

## CYLINDER HEAD GASKETS

The cylinder head gasket is a multi-layer type. Only one thickness of gasket is available.

## CAMSHAFT POSITION SENSOR

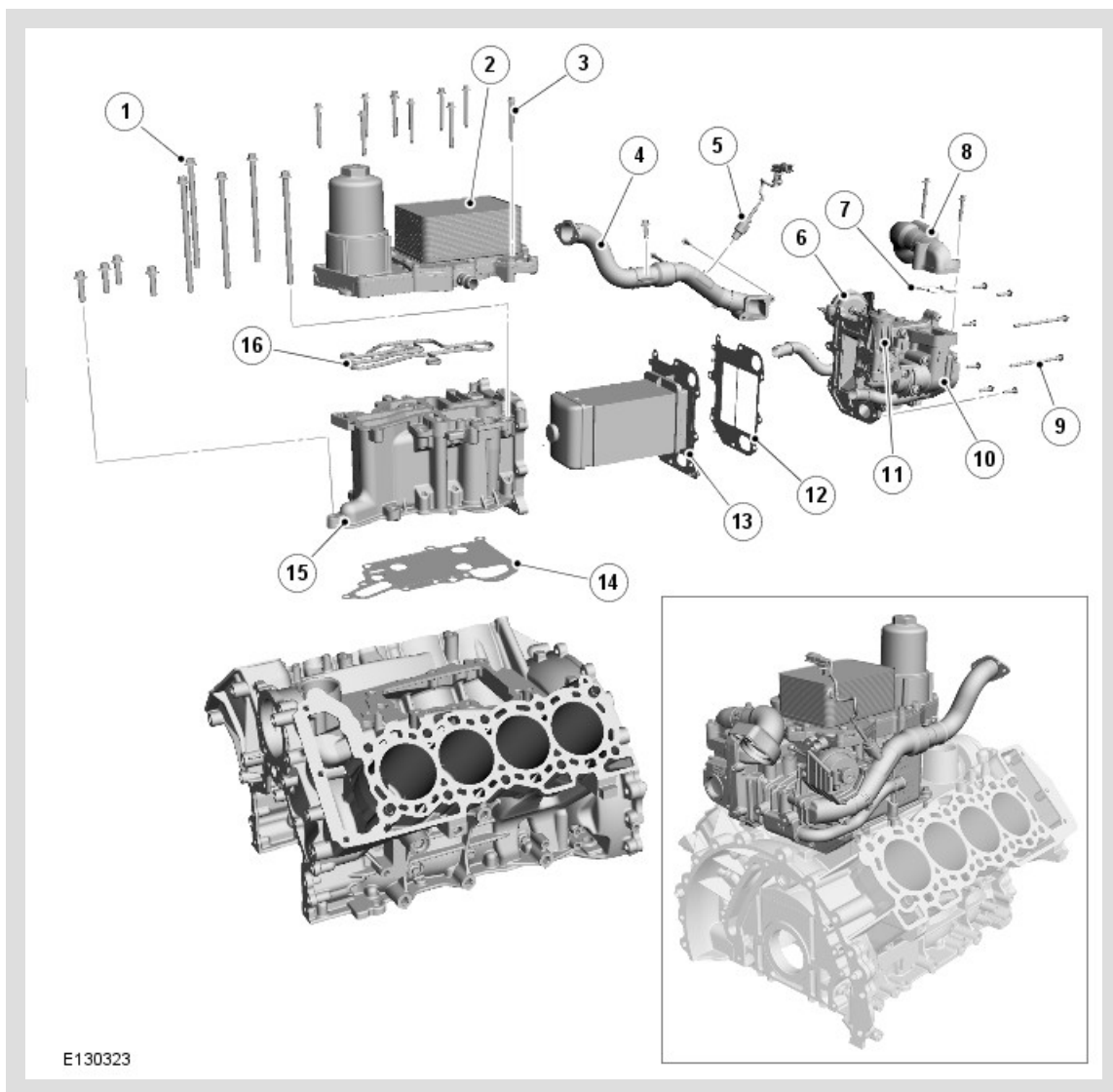


| ITEM | DESCRIPTION              |
|------|--------------------------|
| 1    | Camshaft position sensor |
| 2    | O-ring seal              |
| 3    | Intake camshaft          |
| 4    | Bolt (1 off)             |
| 5    | Left cylinder head       |

The CMP (camshaft position) sensor locates through a hole in a flange on the rear right side of the left cylinder head. The rear of the intake camshaft incorporates a reluctor ring, which is used in conjunction with the sensor to measure engine position.

For additional information, refer to: [Electronic Engine Controls](#) (303-14E Electronic Engine Controls - TDV8 4.4L Diesel, Description and Operation).

## EXHAUST GAS RECIRCULATION VALVE AND COOLER ASSEMBLY



| ITEM | DESCRIPTION                                            |
|------|--------------------------------------------------------|
| 1    | Bolt (9 off)                                           |
| 2    | Engine oil cooler and filter assembly (reference only) |
| 3    | Bolt (10 off) (reference only)                         |
| 4    | Exhaust gas recirculation outlet pipe                  |
| 5    | Exhaust gas recirculation temperature sensor           |
| 6    | Bypass valve vacuum actuator                           |
| 7    | Gasket                                                 |

|    |                                                                |
|----|----------------------------------------------------------------|
| 8  | Exhaust gas recirculation inlet pipe from exhaust manifold     |
| 9  | Bolt (4 off)                                                   |
| 10 | Exhaust gas recirculation valve and bypass assembly            |
| 11 | Exhaust gas recirculation valve motor                          |
| 12 | Gasket                                                         |
| 13 | Exhaust gas recirculation cooler tube                          |
| 14 | Exhaust gas recirculation cooler gasket                        |
| 15 | Exhaust gas recirculation cooler end tank housing              |
| 16 | Gasket - Engine oil cooler to exhaust gas recirculation cooler |

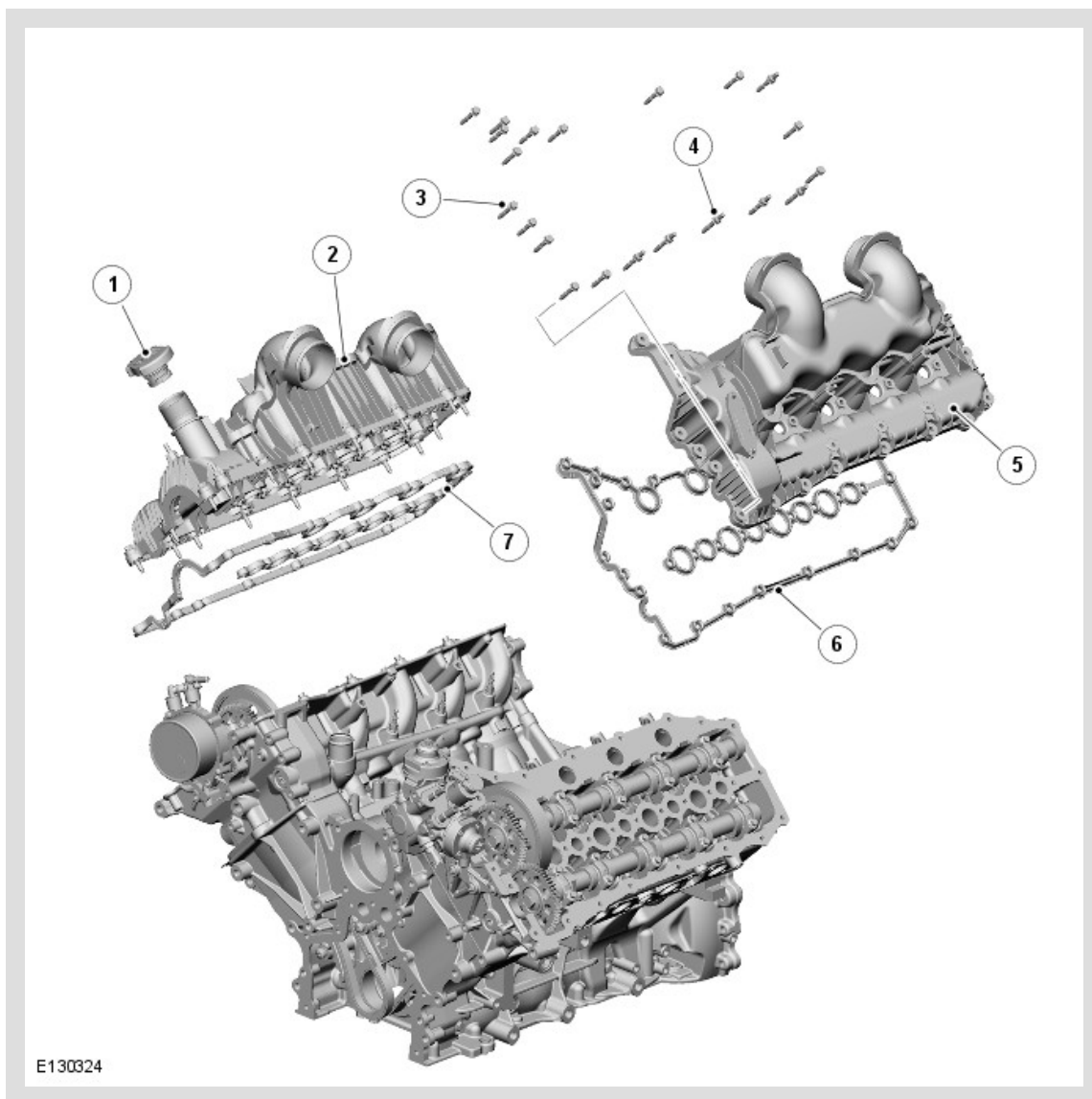
The EGR valve and bypass assembly and the EGR cooler are a combined unit located on the top of the engine crankcase, in the vee between the cylinder heads.

The unit comprises a cooler assembly which uses a cooler housing to circulate engine coolant around an EGR cooler tube which is located within the housing. The tube is sealed to the housing with a gasket and secured with bolts.

The EGR valve and bypass assembly is located on the end of the cooler tube and is sealed with a gasket. The assembly comprises the valve housing to which is attached a vacuum operated bypass valve actuator and the EGR valve motor.

For additional information, refer to: [Engine Emission Control](#) (303-08E Engine Emission Control - TDV8 4.4L Diesel, Description and Operation).

## CAMSHAFT COVERS



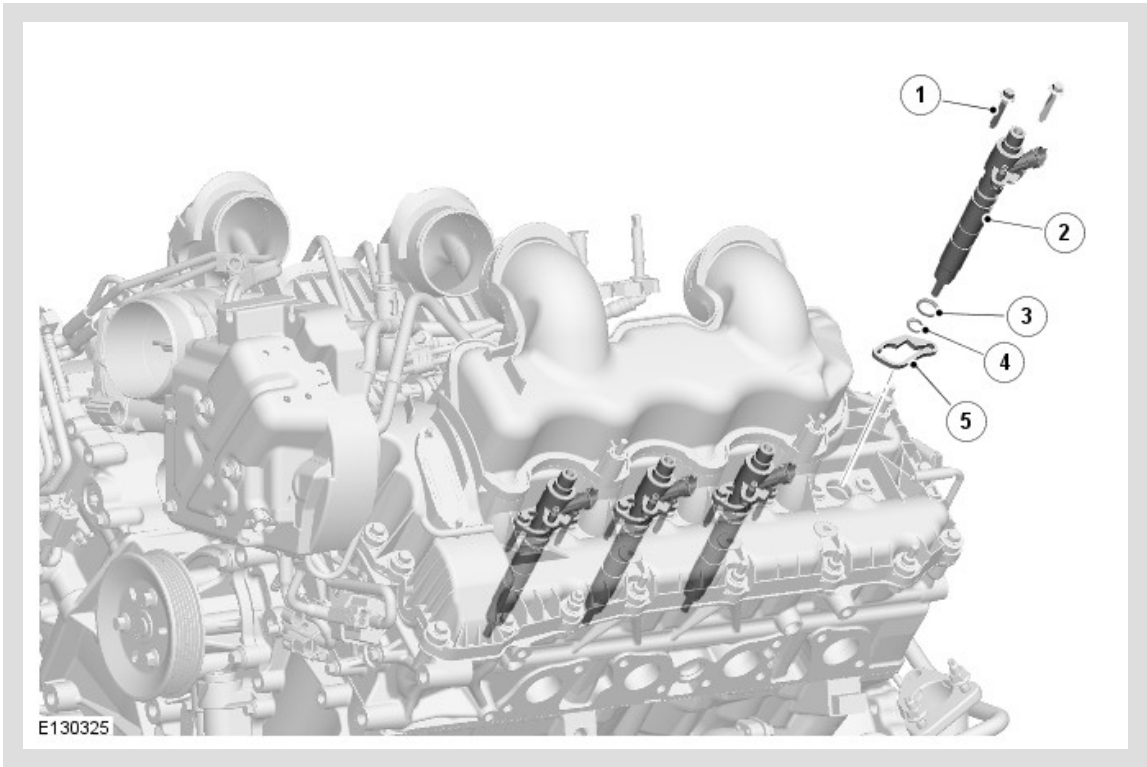
| ITEM | DESCRIPTION                                                           |
|------|-----------------------------------------------------------------------|
| 1    | Oil filler cap (right camshaft cover)                                 |
| 2    | Camshaft cover - right                                                |
| 3    | Bolt (16 off left camshaft cover - 10 off right camshaft cover)       |
| 4    | Stud and nut (7 off left camshaft cover - 8 off right camshaft cover) |
| 5    | Camshaft cover - left                                                 |
| 6    | Manifold gasket - left                                                |
| 7    | Manifold gasket - right                                               |

The intake manifolds are integrated with the camshaft covers. The right camshaft cover incorporates the oil filler cap assembly.

FUEL INJECTORS

 **NOTE:**

Left cylinder head and camshaft cover shown, right similar.



| ITEM | DESCRIPTION                           |
|------|---------------------------------------|
| 1    | Bolt (2 off per injector)             |
| 2    | Injector (8 off, 4 per cylinder head) |
| 3    | O-ring seal                           |
| 4    | Sealing washer                        |
| 5    | Retaining clamp                       |

The 8, top fed, piezo electrically controlled fuel injectors are mounted in the camshaft covers, which locates them centrally in each cylinder head. A washer and an O-ring seals each injector to the manifold interface. Each injector is retained in position with a retaining clamp which is secured to the base part of the manifold with 2 bolts.

The injectors are protected by sound absorbing covers.

The start of fuel injection and the quantity of fuel injected is controlled directly by the ECM .

For additional information, refer to: [Electronic Engine Controls](#) (303-14E Electronic Engine Controls - TDV8 4.4L Diesel, Description and Operation).

## **LUBRICATION SYSTEM**

The lubrication system is a wet oil pan, pressure fed type. It lubricates the engine sliding surfaces, dissipates heat, absorbs fuel combustion residue and seals off the gap between the cylinder and piston.

Oil is drawn from the reservoir in the oil pan and pressurized by the crankshaft driven oil pump. The output from the oil pump is then filtered and distributed through internal oil passageways.

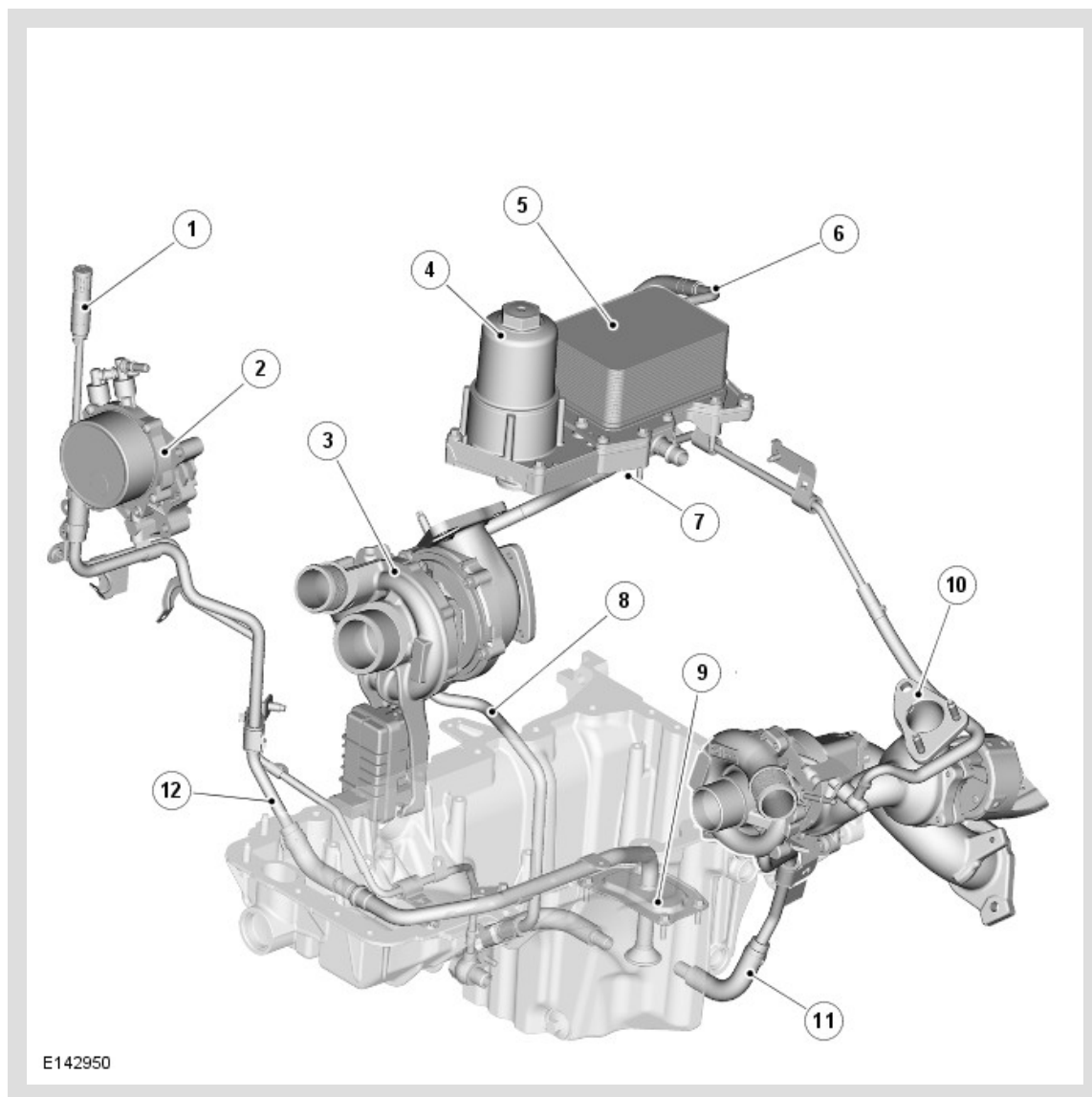
All moving parts are lubricated by pressure or splash oil. Pressurized oil is also provided for operation of the hydraulic lash adjusters and the timing gear chain tensioners.

The engine is lubricated by a force-fed oil circulation system with a full flow oil filter. The oil cooler forms a unit with the oil filter, which is mounted centrally in the middle of the cylinder block between the 2 banks of cylinders. The engine oil is cooled using the engine cooling system. This eliminates the need for an additional engine oil cooler.

Oil returns to the oil pan under gravity. Large drain holes through the cylinder heads and cylinder block ensure the quick return of the oil, reducing the volume of oil required and enabling an accurate check of the contents soon after the engine stops.

System replenishment is through the oil filler cap on the right camshaft cover.

## **TURBOCHARGER OIL RETURN**



| ITEM | DESCRIPTION                        |
|------|------------------------------------|
| 1    | Oil extraction tube                |
| 2    | Vacuum and oil scavenge pump       |
| 3    | Variable geometry turbocharger     |
| 4    | Engine oil filter (ref)            |
| 5    | Engine oil cooler                  |
| 6    | Oil feed pipe to turbochargers     |
| 7    | Cooler and filter housing assembly |

|    |                           |
|----|---------------------------|
| 8  | Oil return pipe           |
| 9  | Secondary reservoir cover |
| 10 | Fixed vane turbocharger   |
| 11 | Oil return pipe           |
| 12 | Oil scavenge pipe         |

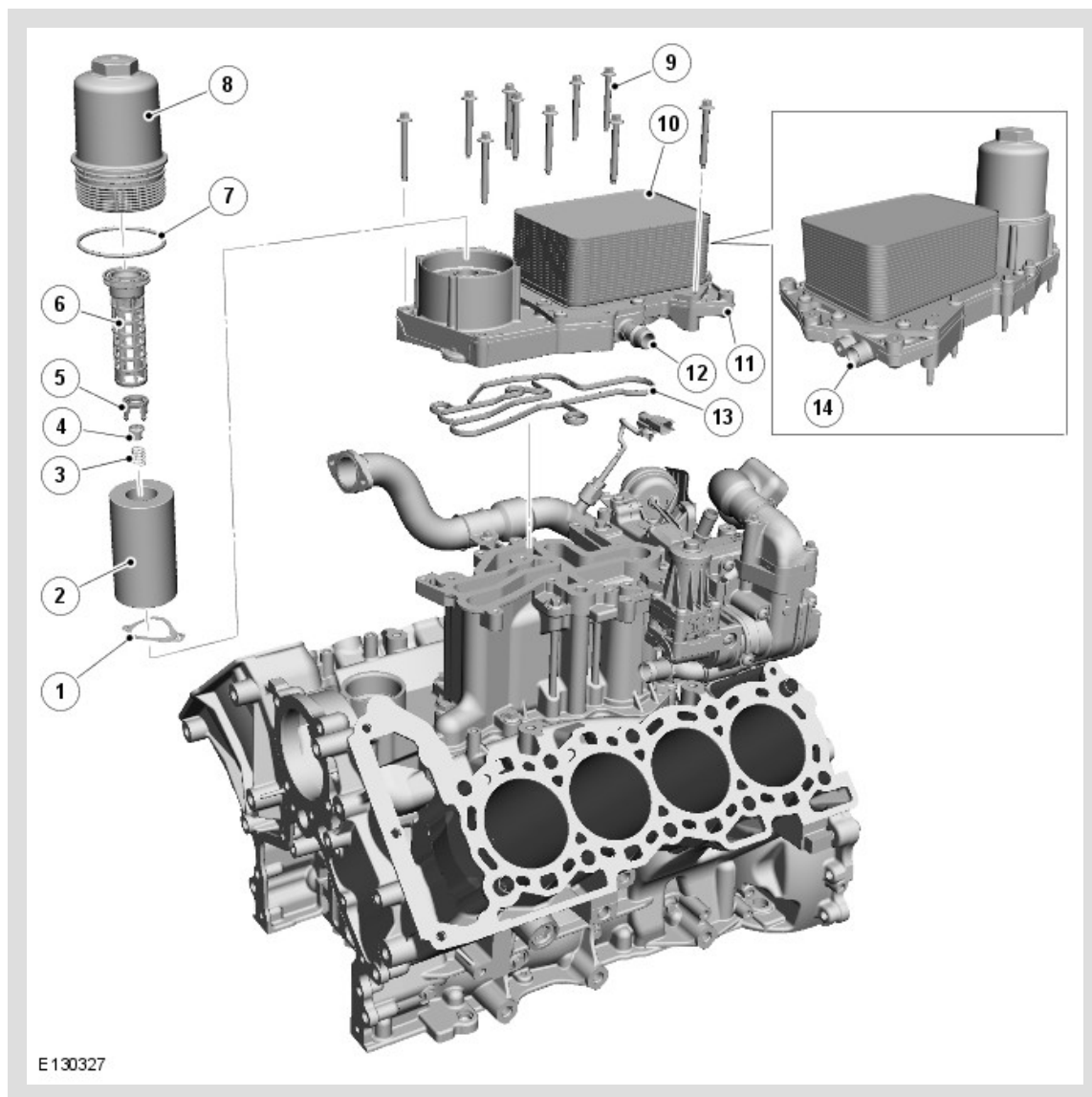
Due to the wide range of inclines Land Rover vehicles operate across, the geometry of the oil pan has been configured to guarantee oil pick-up across all operating angles. An oil scavenge system has been developed to guarantee excellent oil flow through the turbochargers on severe side-slopes. At extreme angles there is a risk the turbochargers may be below the oil level in the oil pan, restricting oil return flow. To overcome this, the engine uses:

- The oil pump to supply oil to the turbocharger bearings from the main reservoir of the oil pan
- A secondary reservoir in the oil pan, sealed off from the main reservoir, to receive oil draining from the turbocharger bearings
- A scavenge pump, integrated into the vacuum pump, to assist the flow of oil draining from the turbocharger bearings by scavenging oil from the secondary reservoir into the right camshaft cover.

#### OIL PICK-UP

An oil pick-up assembly is immersed in the bottom of the oil pan to provide a supply to the oil pump during all normal vehicle attitudes. The oil pick-up contains a mesh screen to filter the oil and prevent debris from entering the oil system.

#### OIL FILTER AND OIL COOLER ASSEMBLY



| ITEM | DESCRIPTION        |
|------|--------------------|
| 1    | Leaf spring        |
| 2    | Filter element     |
| 3    | Compression spring |
| 4    | Valve cone         |
| 5    | Valve seat         |
| 6    | Center pipe        |
| 7    | O-ring seal        |

|    |                                               |
|----|-----------------------------------------------|
| 8  | Filter cover                                  |
| 9  | Bolt (10 off)                                 |
| 10 | Oil cooler                                    |
| 11 | Housing assembly                              |
| 12 | Connection to EGR cooler inlet                |
| 13 | Gasket                                        |
| 14 | Engine oil supply connection to turbochargers |

The engine is lubricated by a force-feed oil circulation system with a full flow oil filter. The oil cooler forms a unit with the oil filter, which is mounted centrally in the middle of the cylinder block between the 2 banks of cylinders.

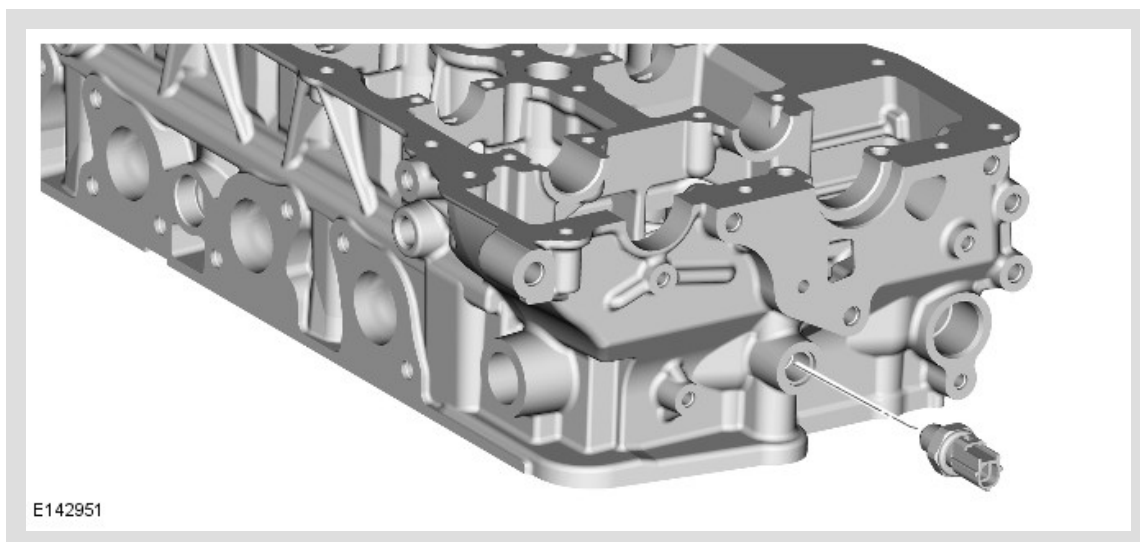
The oil filter is a replaceable cartridge installed on an adapter in the center of the 'vee'. An internal bypass valve permits full flow bypass if the filter becomes blocked.



#### NOTE:

In service care must be taken when removing the oil filter to minimize oil drips and spillage into the engine 'vee' and camshaft covers.

#### OIL PRESSURE SWITCH

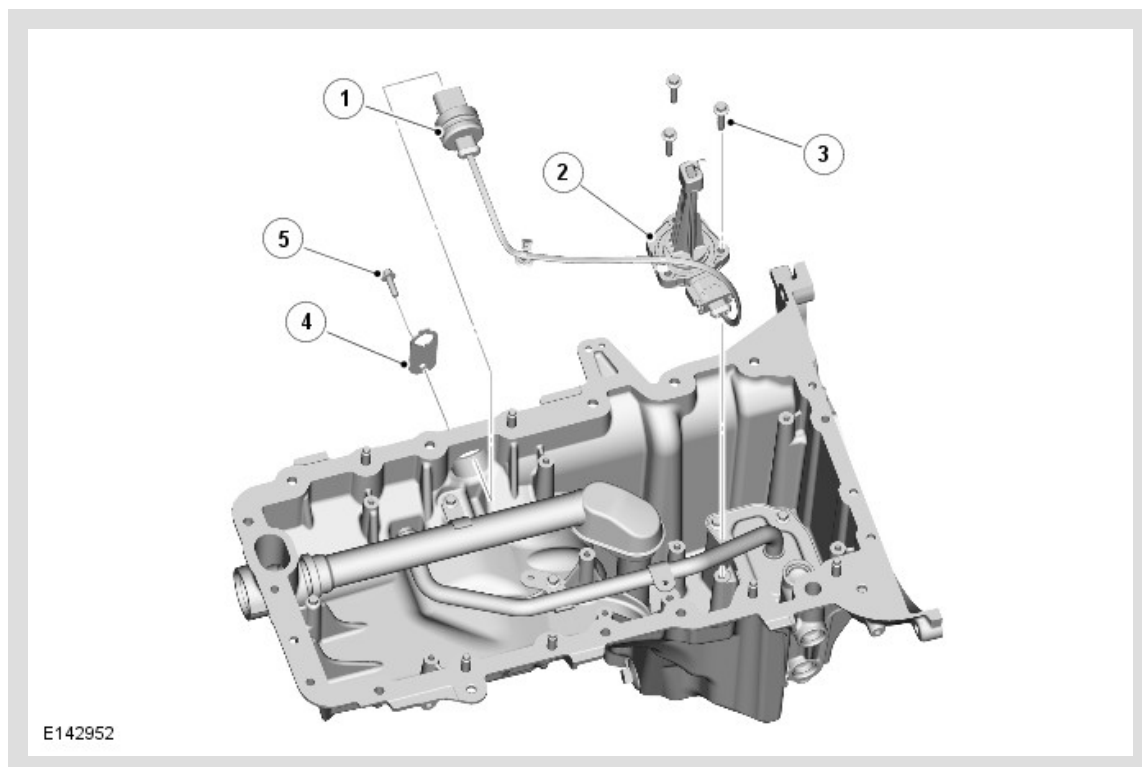


The oil pressure switch is located in the front of the right cylinder head. The switch connects a ground input to the CJB (central junction box) when oil pressure is present.

The switch operates at a pressure of 0.15 to 0.41 bar (2.2 to 5.9 lbf/in<sup>2</sup>).

For additional information, refer to: [Electronic Engine Controls](#) (303-14E Electronic Engine Controls - TDV8 4.4L Diesel, Description and Operation).

#### OIL LEVEL AND TEMPERATURE SENSOR



| ITEM | DESCRIPTION                      |
|------|----------------------------------|
| 1    | Sensor link harness              |
| 2    | Oil level and temperature sensor |
| 3    | Screw (3 off)                    |
| 4    | Connector bracket                |
| 5    | Screw                            |

The oil level and temperature sensor is installed in the bottom of the oil pan and secured with three screws. A link lead connects the sensor to the engine harness at the right side of the oil pan.

The sensor has three connections for a power supply from the ECM relay in the EJB (engine junction box) , and signal lines to the ECM for level and temperature information.

To measure the oil level, the sensor sends an ultrasonic pulse vertically upward. It then measures the time for the pulse to be reflected back from the top surface of the oil.

Warnings are displayed in the message center if the oil level is not maintained within the safe operating levels (minimum and maximum). A warning is also displayed if there is a fault with the oil level monitoring system.

#### OIL LEVEL CHECK

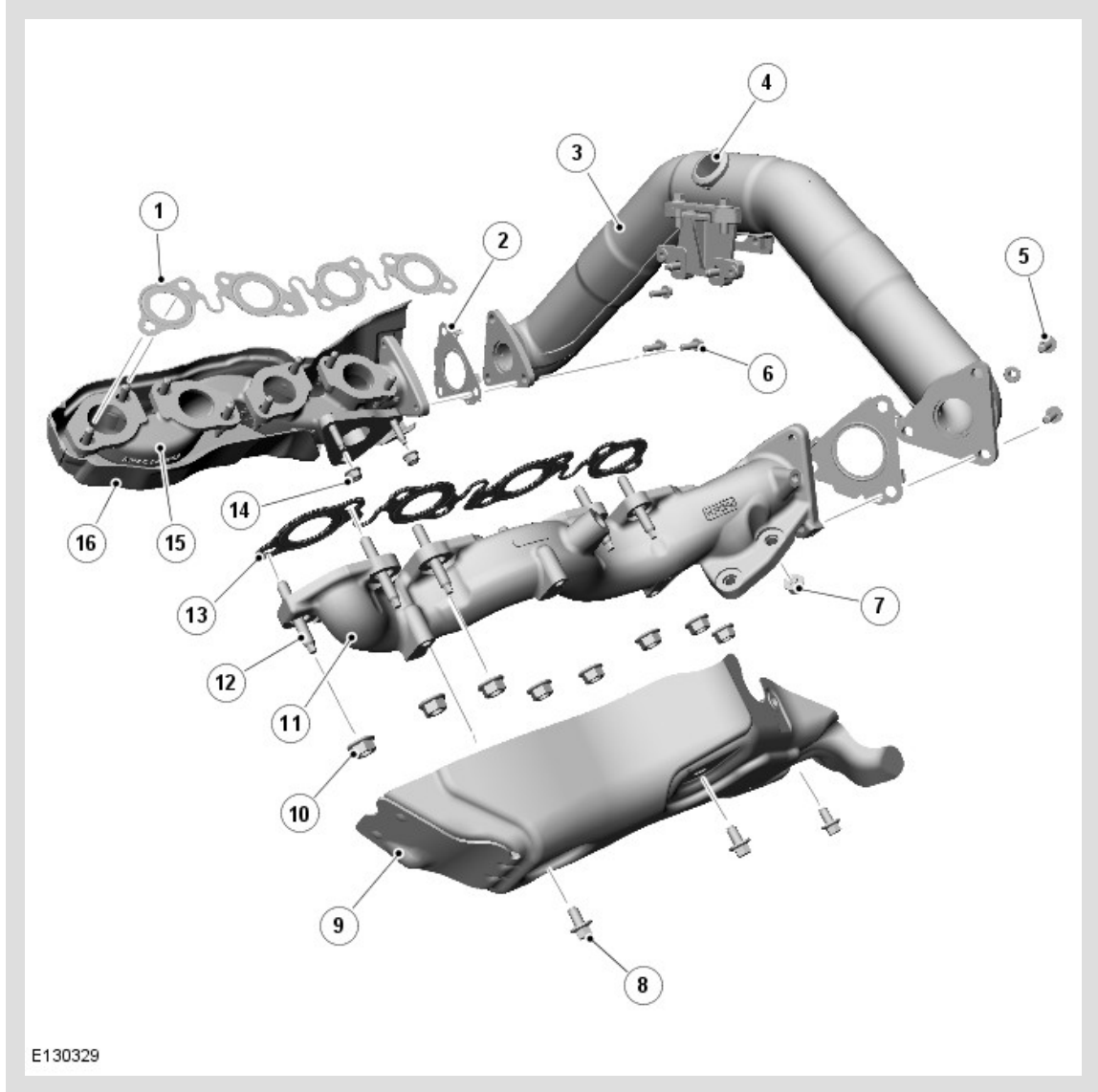
The engine oil level can be displayed in the message center on the instrument cluster.

For accuracy, oil level checks should be performed with the vehicle on level ground when the oil is hot. The vehicle needs to stand for approximately 10 minutes, after the engine is switched off, to allow the oil to drain back into the sump and the oil level to stabilize. The oil level system will not give a reading until the oil level has stabilized.

The method of checking the oil level can be found in the oil draining and filling procedures. For additional information, refer to: (303-01E Engine - TDV8 4.4L Diesel)

[Engine Oil Draining and Filling](#) (General Procedures),  
[Engine Oil Vacuum Draining and Filling](#) (General Procedures).

#### EXHAUST MANIFOLDS



| ITEM | DESCRIPTION                                                                    |
|------|--------------------------------------------------------------------------------|
| 1    | Right exhaust manifold gasket                                                  |
| 2    | Gasket - exhaust manifold to crossover pipe                                    |
| 3    | Crossover pipe                                                                 |
| 4    | Crossover pipe EGR connection to EGR valve and bypass assembly                 |
| 5    | Bolt and nut - crossover pipe to left exhaust manifold (2 off bolt, 1 off nut) |
| 6    | Bolt - crossover pipe to right exhaust manifold (3 off)                        |
| 7    | Nut - exhaust manifold to turbocharger flange (3 off)                          |

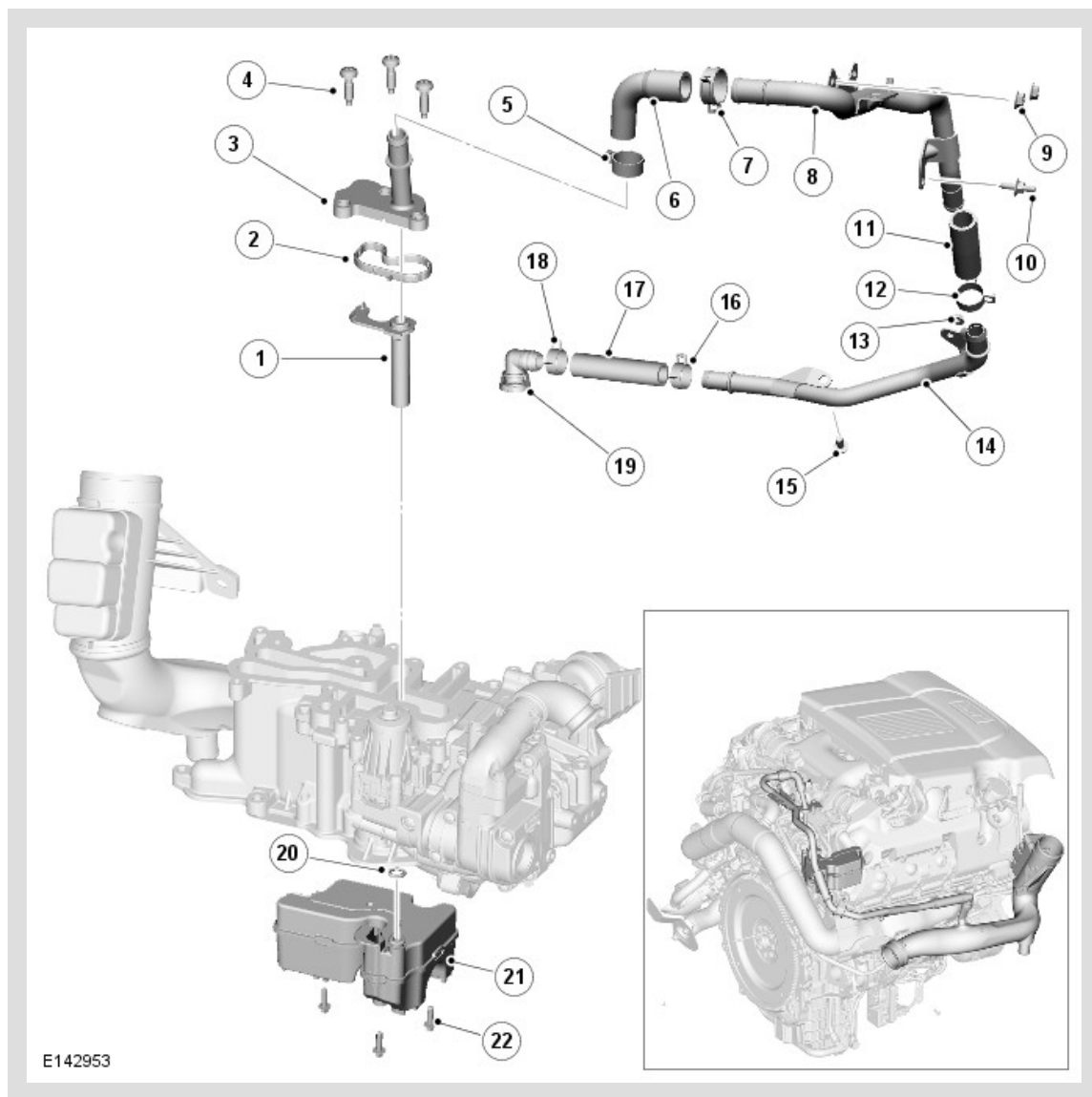
|    |                                                      |
|----|------------------------------------------------------|
| 8  | Bolt - left exhaust manifold heat shield (3 off)     |
| 9  | Left exhaust manifold heat shield                    |
| 10 | Nut - exhaust manifold to cylinder head stud (8 off) |
| 11 | Left exhaust manifold                                |
| 12 | Stud (8 off)                                         |
| 13 | Left exhaust manifold gasket                         |
| 14 | Nut - right exhaust manifold to turbocharger (3 off) |
| 15 | Right exhaust manifold                               |
| 16 | Right exhaust manifold heat shield                   |

The exhaust manifolds are handed for each cylinder bank. Each manifold is secured to its respective cylinder head with 8 studs and 8 flanged nuts. A gasket seals the manifold to the cylinder head.

Each exhaust manifold has 2 flanges. One flange at the rear of the manifold provides the mounting for the crossover pipe which is sealed with a gasket and secured with 2 bolts and 1 stud and a flanged nut on the left manifold and 3 bolts on the right manifold. The second flange faces downwards and provides the attachment for the turbochargers. Each turbocharger is secured to its manifold with 3 studs and 3 flanged nuts.

Each manifold is fitted with a heat shield. The heat shield is secured to the manifold with 3 bolts which screw into threaded holes in the manifold.

## CRANKCASE VENTILATION



| ITEM | DESCRIPTION        |
|------|--------------------|
| 1    | Baffle             |
| 2    | Gasket             |
| 3    | Crankcase vent cap |
| 4    | Screw (3 off)      |
| 5    | Hose clip          |
| 6    | Hose               |
| 7    | Hose clip          |

|    |                                  |
|----|----------------------------------|
| 8  | Vent tube                        |
| 9  | Nut (2 off)                      |
| 10 | Stud and nut                     |
| 11 | Short hose                       |
| 12 | Hose clip                        |
| 13 | Nut                              |
| 14 | Tube                             |
| 15 | Bolt                             |
| 16 | Hose clip                        |
| 17 | Long hose                        |
| 18 | Hose clip                        |
| 19 | Quick fit connector              |
| 20 | O-ring seal                      |
| 21 | Crankcase oil separator assembly |
| 22 | Bolt (3 off)                     |

The crankcase ventilation system ensures that all gasses emitted from the crankcase during engine running are separated from any oil particles and recirculated via the clean air induction system.

Crankcase gasses are drawn into the oil separator unit from the crankcase by a vacuum created by a connection into the air induction system.

The crankcase gasses are circulated around the oil separator where the gas and oil are separated. The gas is returned to the inlet side of the air induction system prior to the primary turbocharger. A series of pipes and hoses route the gasses from the separator to the inlet pipe of the primary turbocharger. The collected oil is drained down from 2 umbrella valves in the base of the separator to the sump.

For additional information, refer to: [Engine Emission Control](#) (303-08E Engine Emission Control - TDV8 4.4L Diesel, Description and Operation).

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## OPERATION

Operation of the engine is controlled by the ECM .

For additional information, refer to: [Electronic Engine Controls](#) (303-14E Electronic Engine Controls - TDV8 4.4L Diesel, Description and Operation).

